Sri Lanka Institute of Information Technology
IE2012: System and Network Programming
Year 2 Semester 1

Assignment

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MLB_WD_CS_0101

Contents

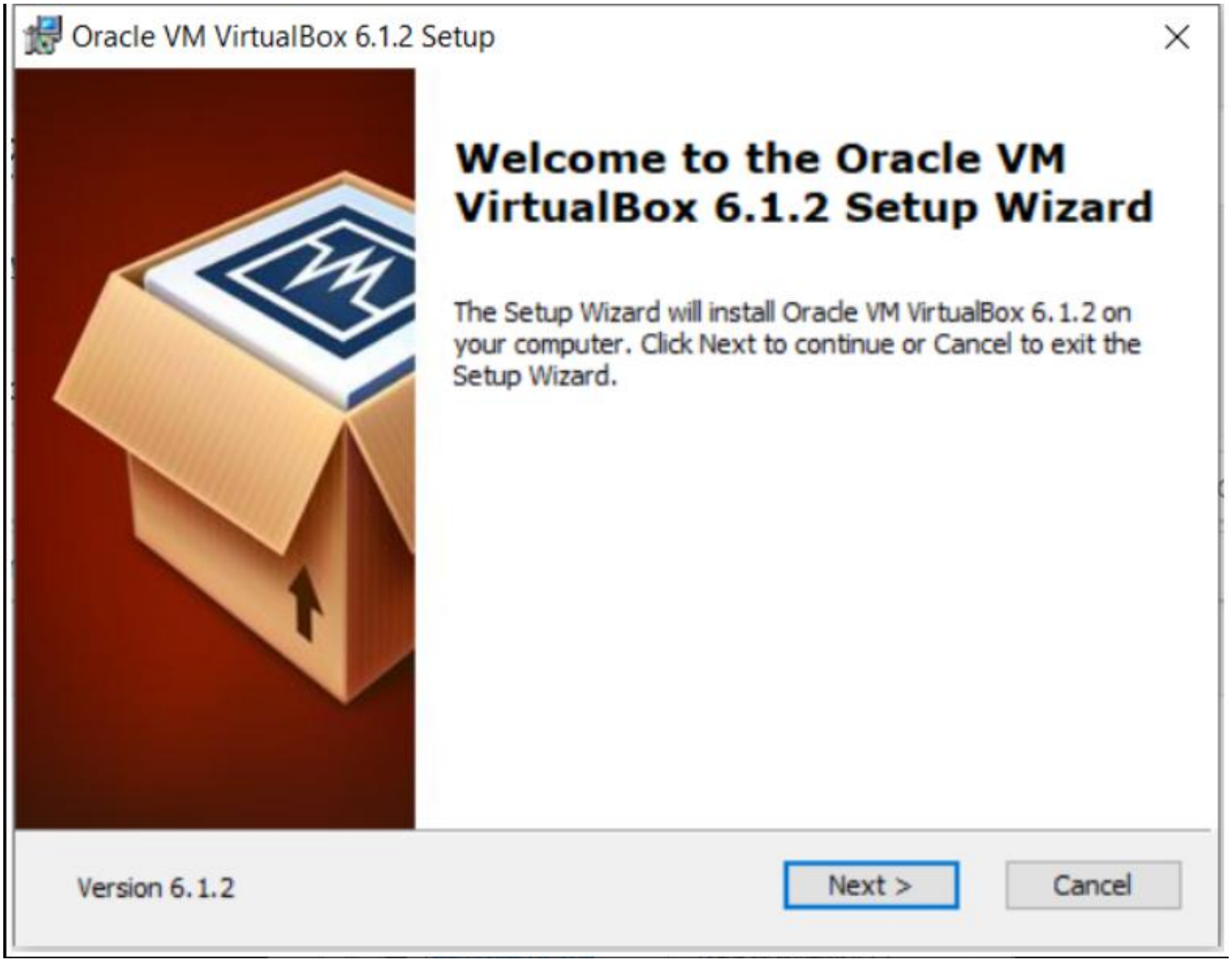
1. Basics of the Linux environment

I. Virtual Machine Setup

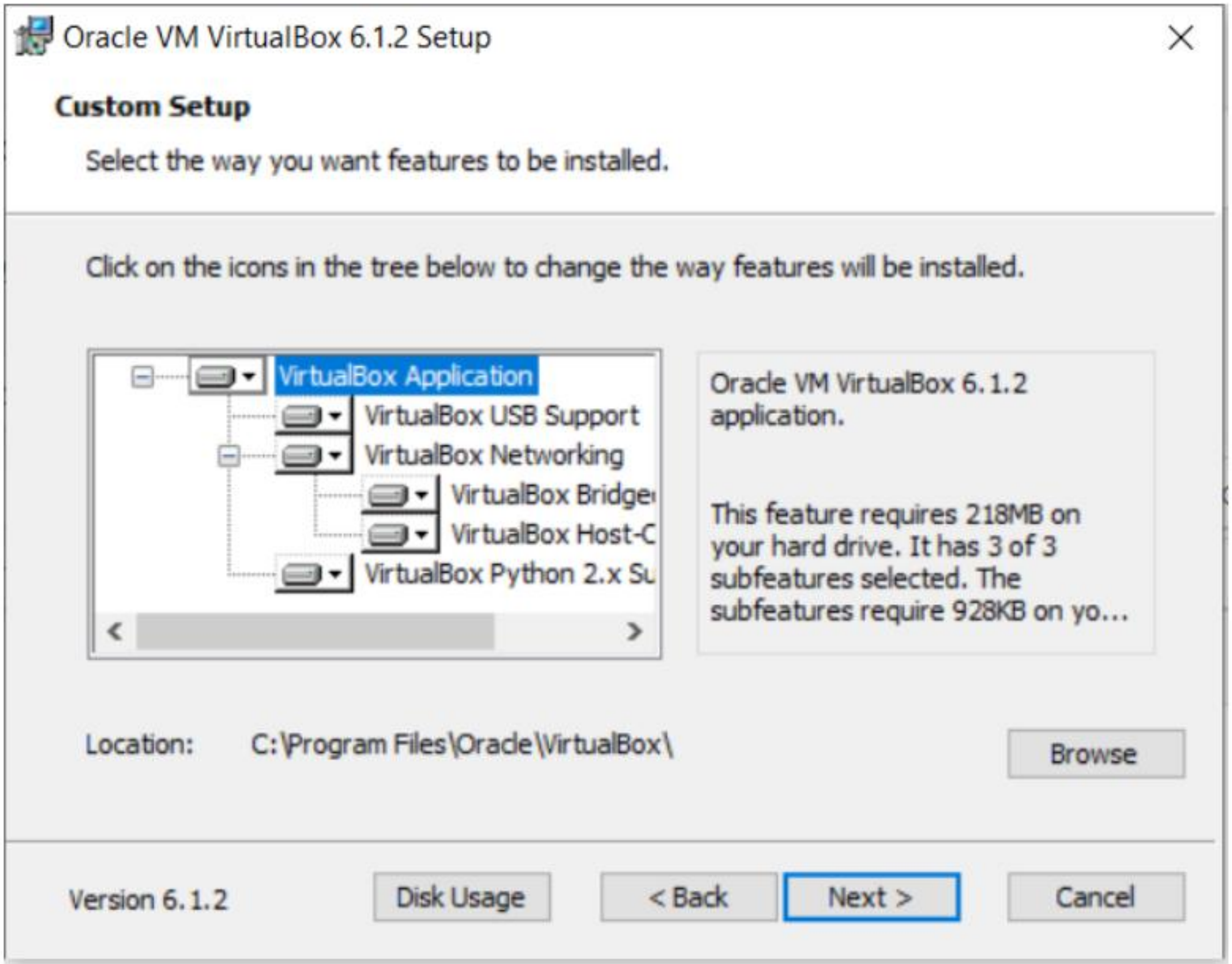
Step 01

Download a virtual emulator for Windows, Linux, or Mac to start. To download VirtualBox, go to the official site **virtualbox.org** and download the latest version for windows.

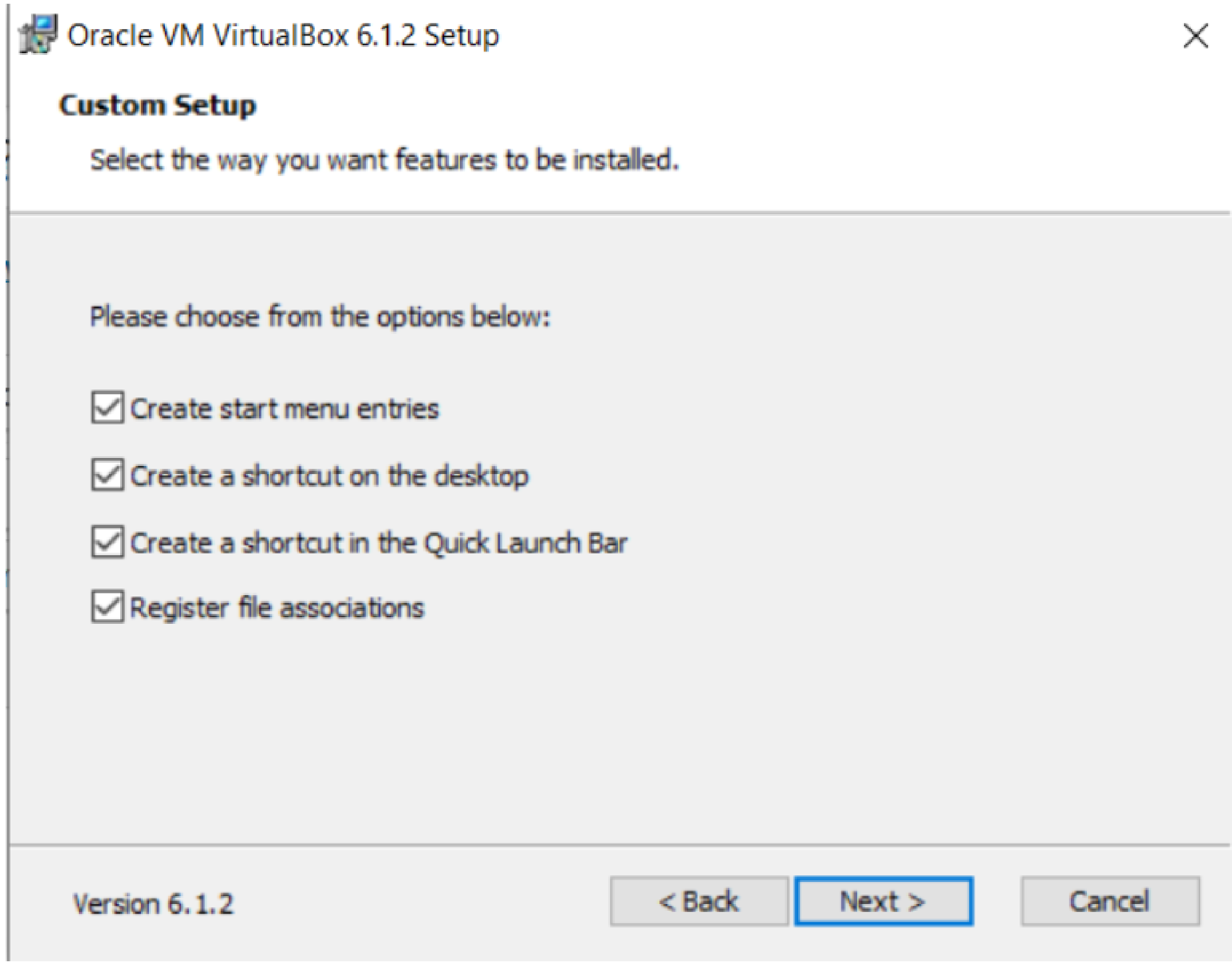
- **Getting Started:**



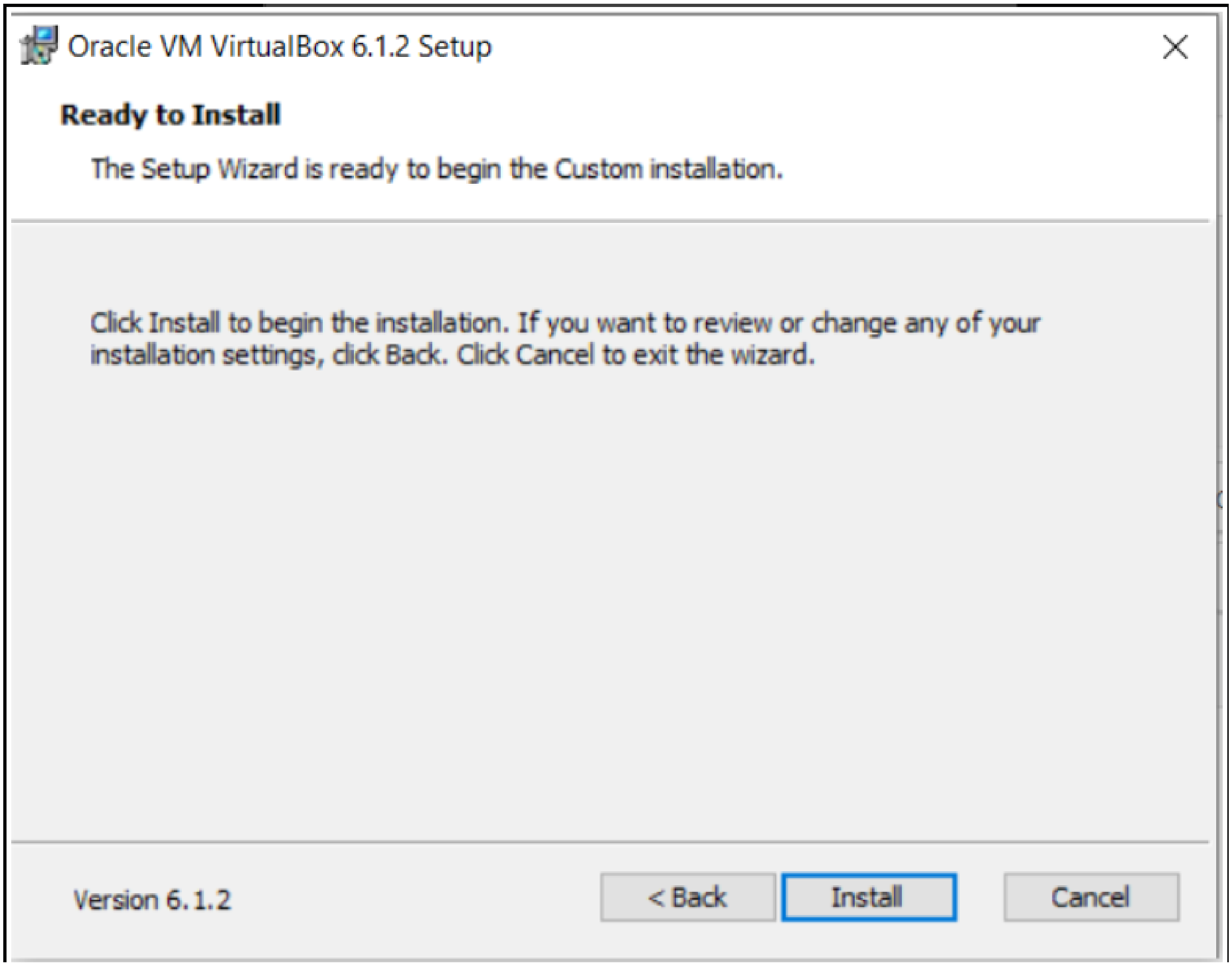
- **Select Installation Location:**



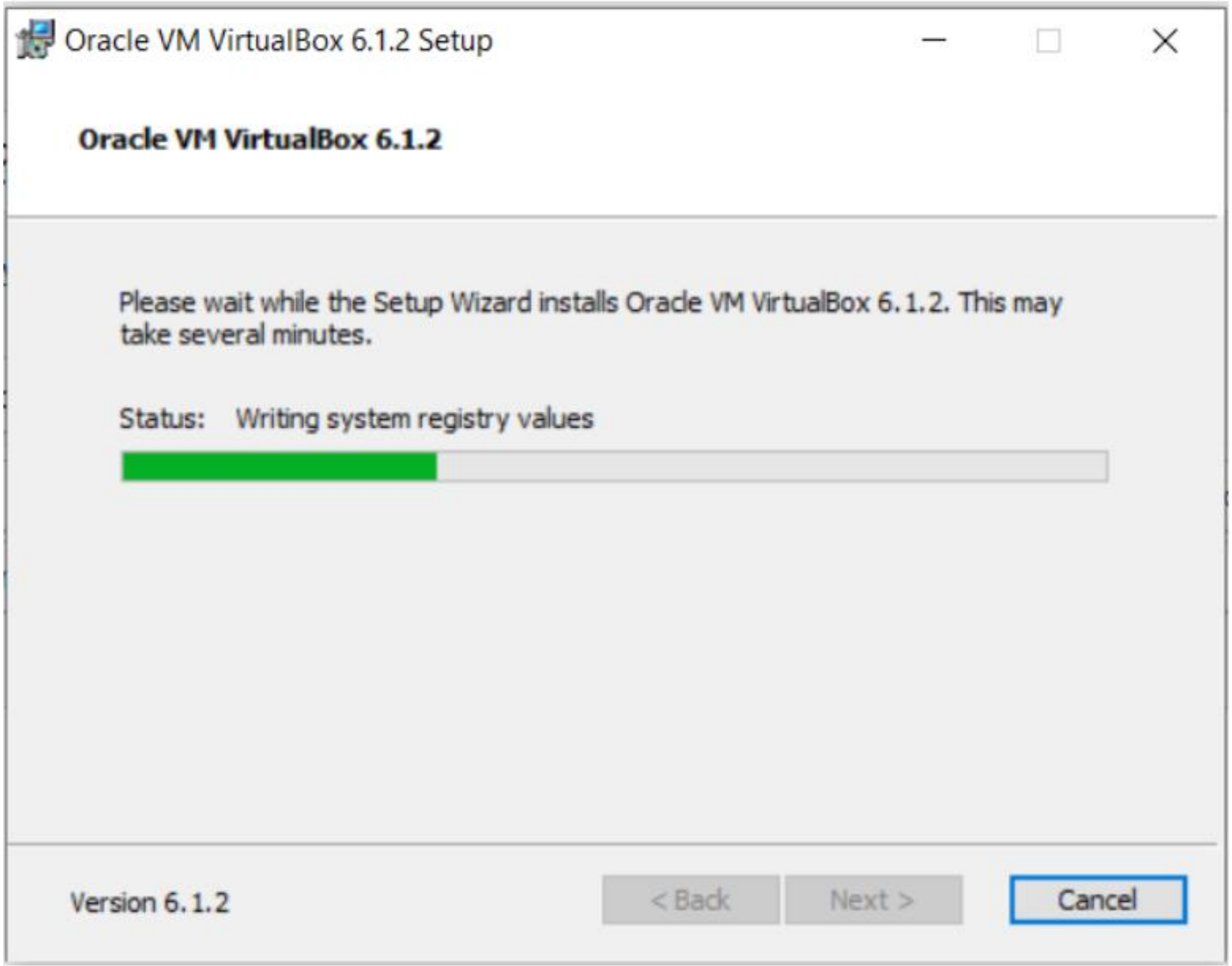
- **Creating Entries and Shortcuts:**



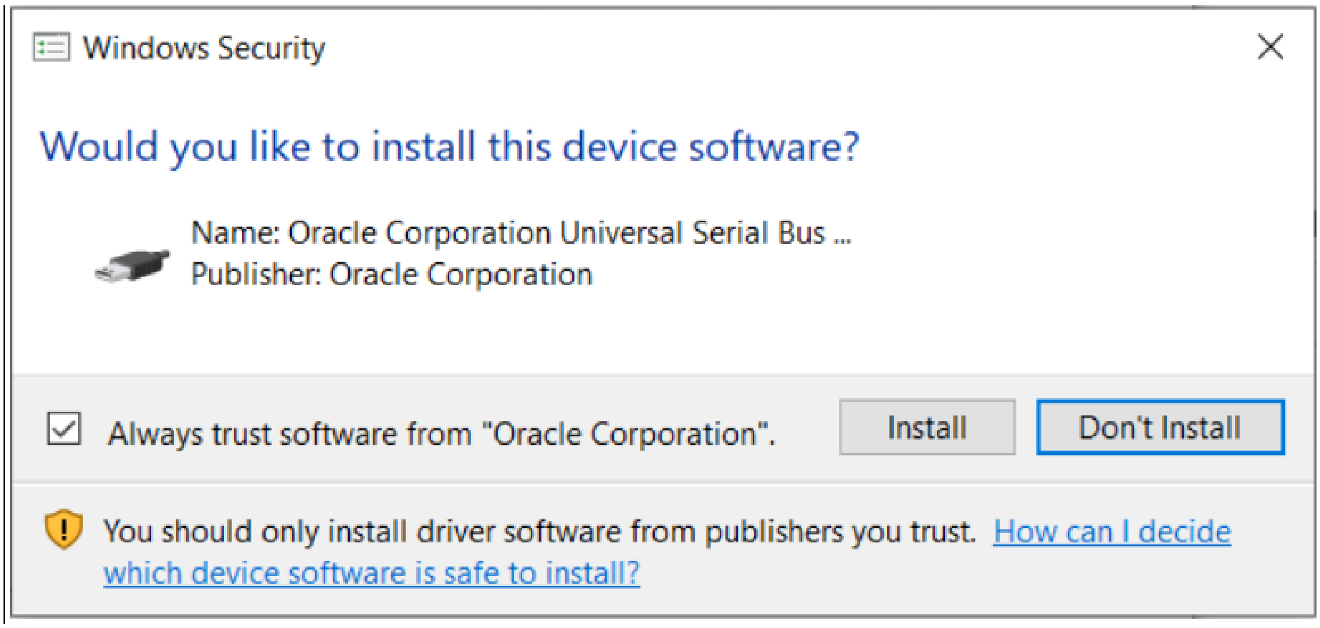
- **Ready to Install:**



- **Installing Files and packages:**

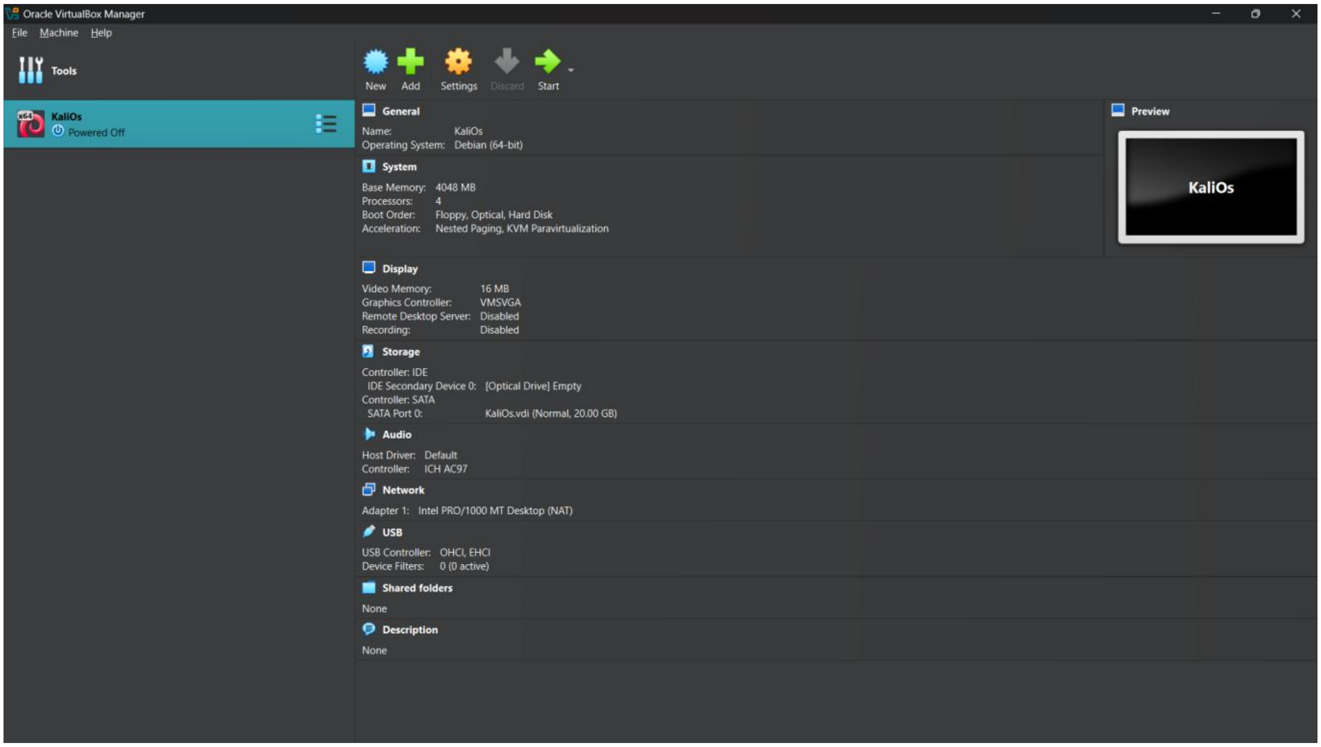


- **Installing Certificates:**



- **Finished Installation:**

We can now continue with the installation (ensure you have Microsoft Visual C++ 2019 redistribution or higher version installed). Once the executable file is run and the administrator privilege supplement pop-up is allowed, we should choose the location and mode of Virtual Box installation. This box will appear when we start Virtual Box via the new desktop shortcut icon once the installation process has finished.



We can select the “New” option in the upper left corner and enter the information for the VM. We need to provide it a name, install location, and installation media image file in order of precedence. The other fields will also be filled automatically with the selected ISO file as soon as you select it. The Linux operating system will need to be installed manually if the check box "Skip Unattended Installation" is selected; otherwise, it will install automatically without our intervention. We can then select the next button by clicking.

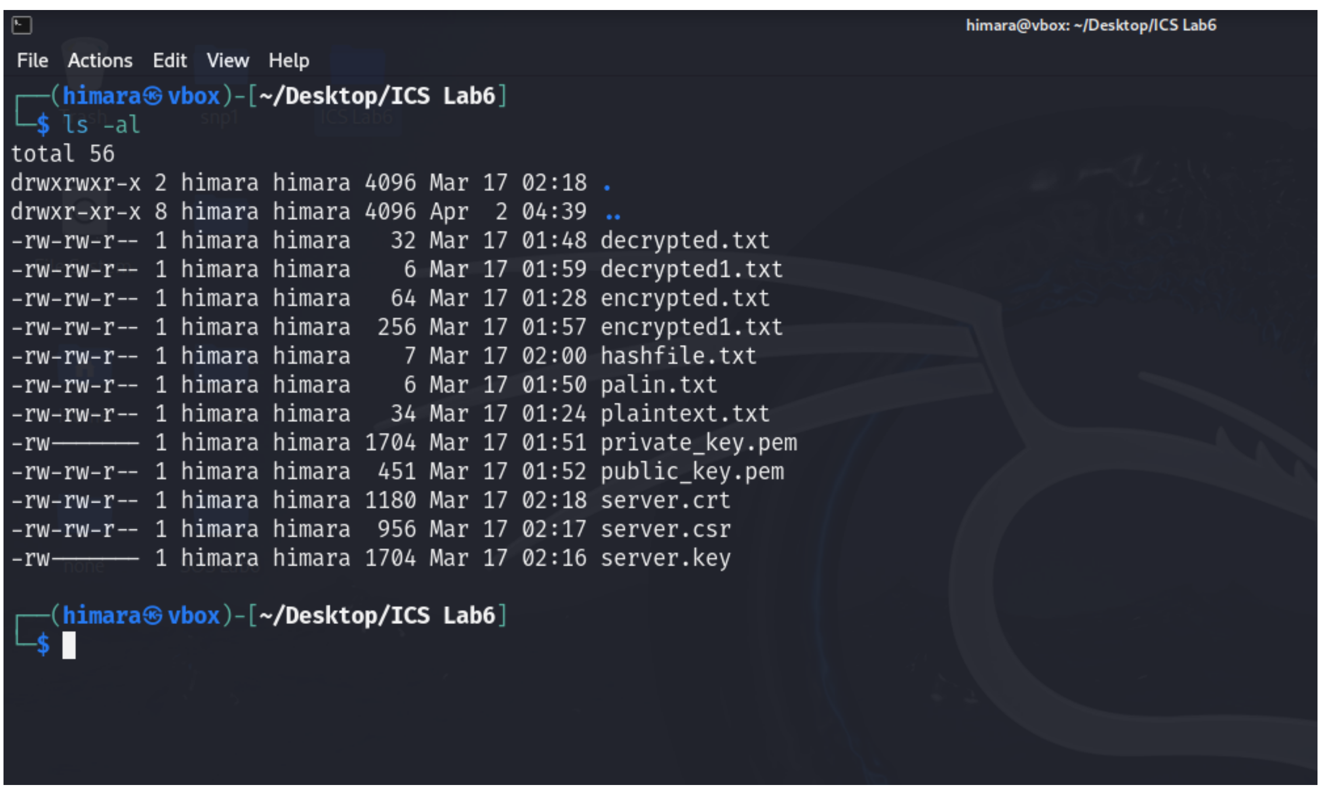
II. Command Line Introduction

1. **ls – Listing files:** This command is used to list the files and directories stored in the current directory. When you type ls and press Enter, the current directory's contents may be viewed as a list.

Other options available:

a) ***– l*** : Long format which shows file details like permissions, size and modification time.

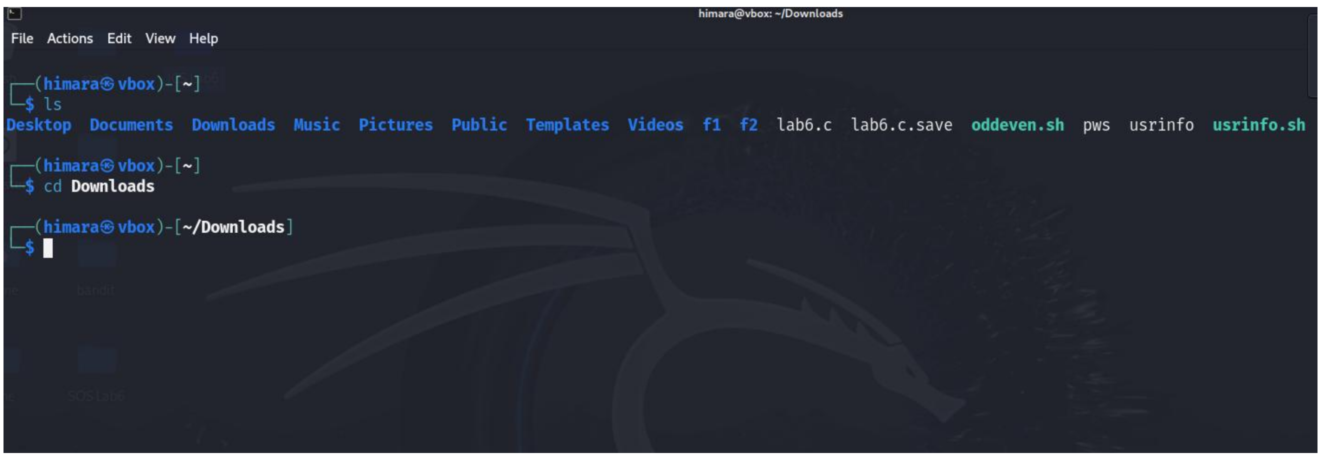
b) ***-a :*** Lists all files, including hidden ones (files that start with a dot).



- You have, from left to right:

- the file permissions (and if your system supports ACLs, you get an ACL flag as well)
- the number of links to that file
- the owner of the file
- the group of the file
- the file size in bytes
- the file modified datetime
- the file name

2. **cd – Change directory:** This Using the command-line interface, it is used to modify the current working directory. If you want to get into a directory called "Downloads" for instance, type cd Downloads.



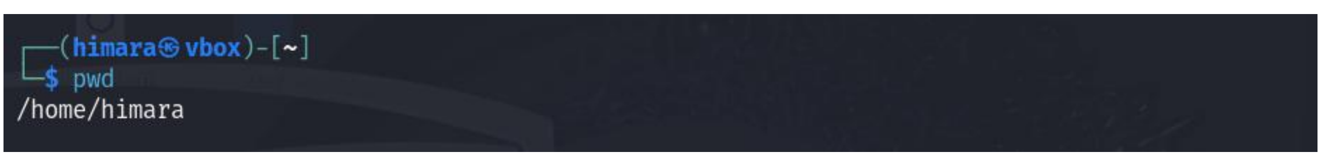
To move back a directory, we can use: ***..cd***

- Examples:

cd /home/user/Downloads: to move to the Downloads directory.

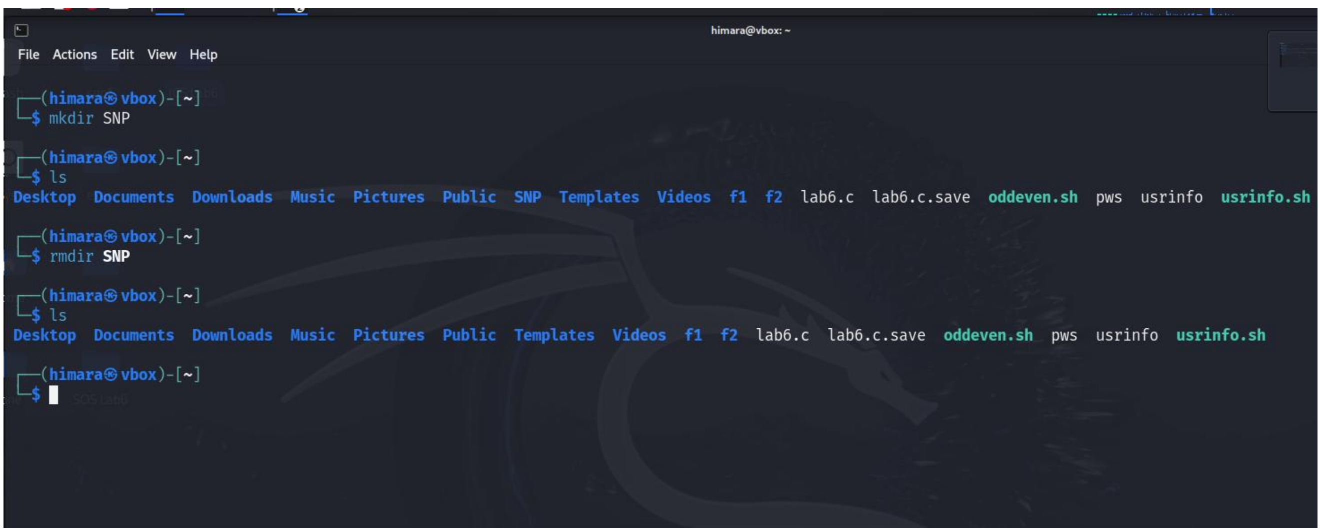
cd .. :to move up one directory level.

3. **pwd – present working directory:** Show your current full path that you in and it helps to you know where you are in the file system.



4. **mkdir – Make Directory:** This command use to create new folders or directories

5. **rmdir – Remove Directory:** This command is used to remove folders or directories, but it can only remove them if they are empty.

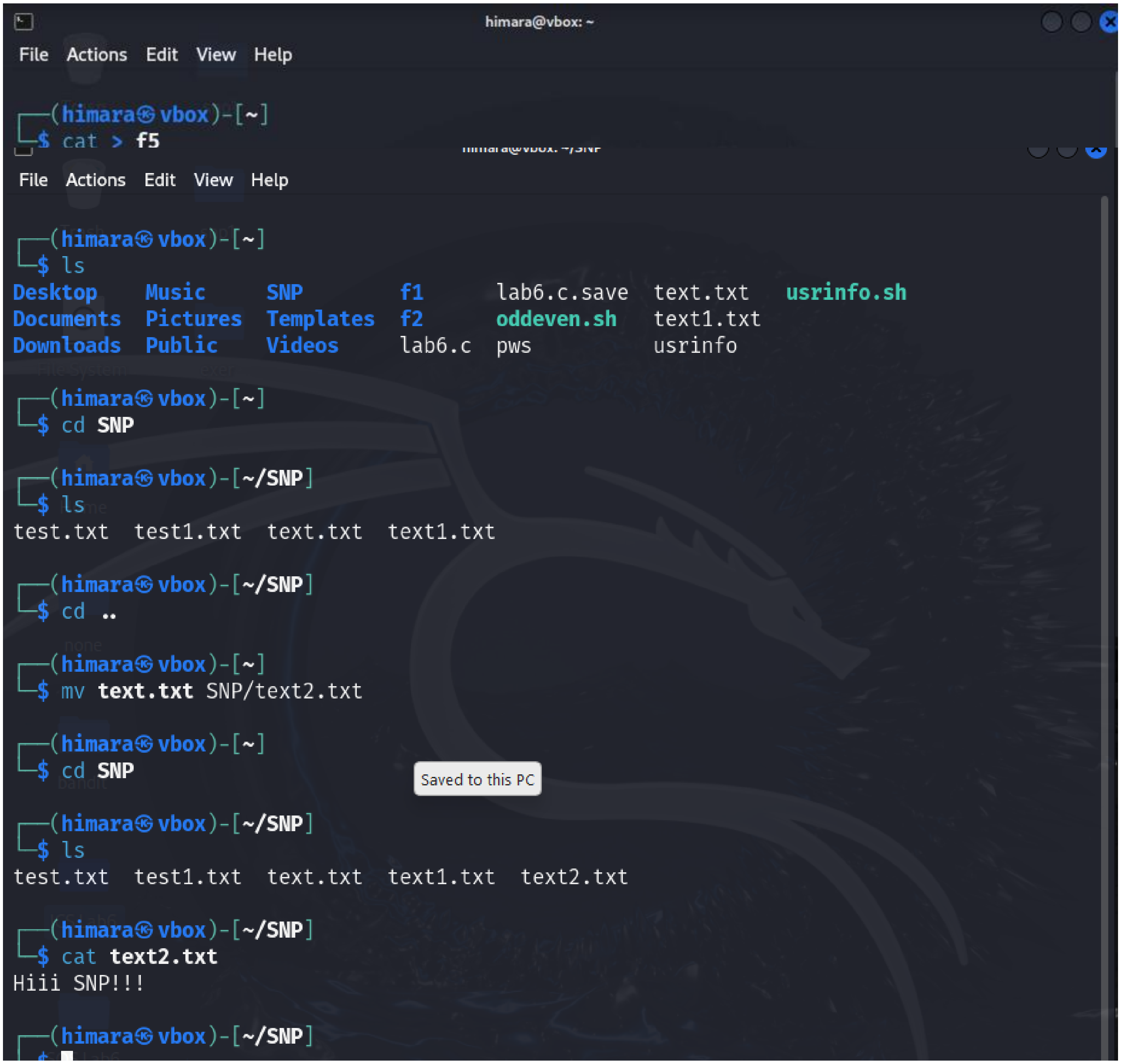


6. **rm -r:** If the folder or directory is not empty you can use this command.Be careful as this command does not ask for confirmation, and it will immediately remove anything you ask it to remove. There is no bin when removing files from the command line, and recovering lost files can be hard
   - rm → remove
   - -r → recursively delete **everything inside** the folder

7. **cp – Copy files/folders:** Copy the contents of source file to the destination file called destination-file. For example, to copy a file named "test. txt" from a folder "SNP" to a file named "test1. txt".

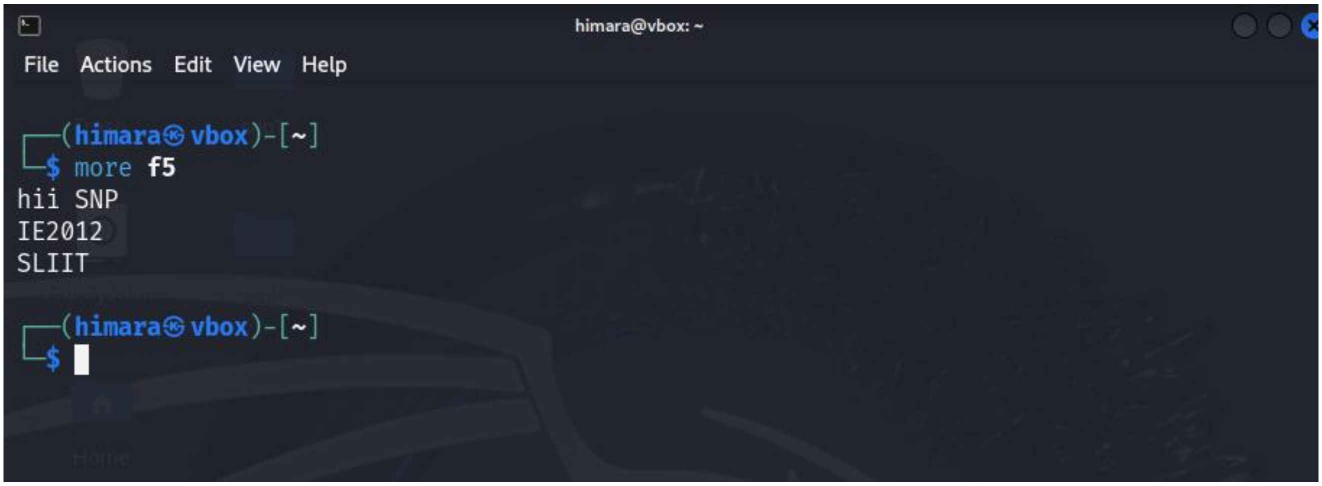
8. **mv – Move or Rename:** This command is used to move files/directories from one location to another, or to rename them.Simple syntax is **mv source_file destination_file to move a file.**

As an example, if we want to transfer a file called "test. txt" to a directory named "SNP" you would use: test1.txt
SNP/test2.txt . To change a name to a file, you would use mv old_filename new_filename.



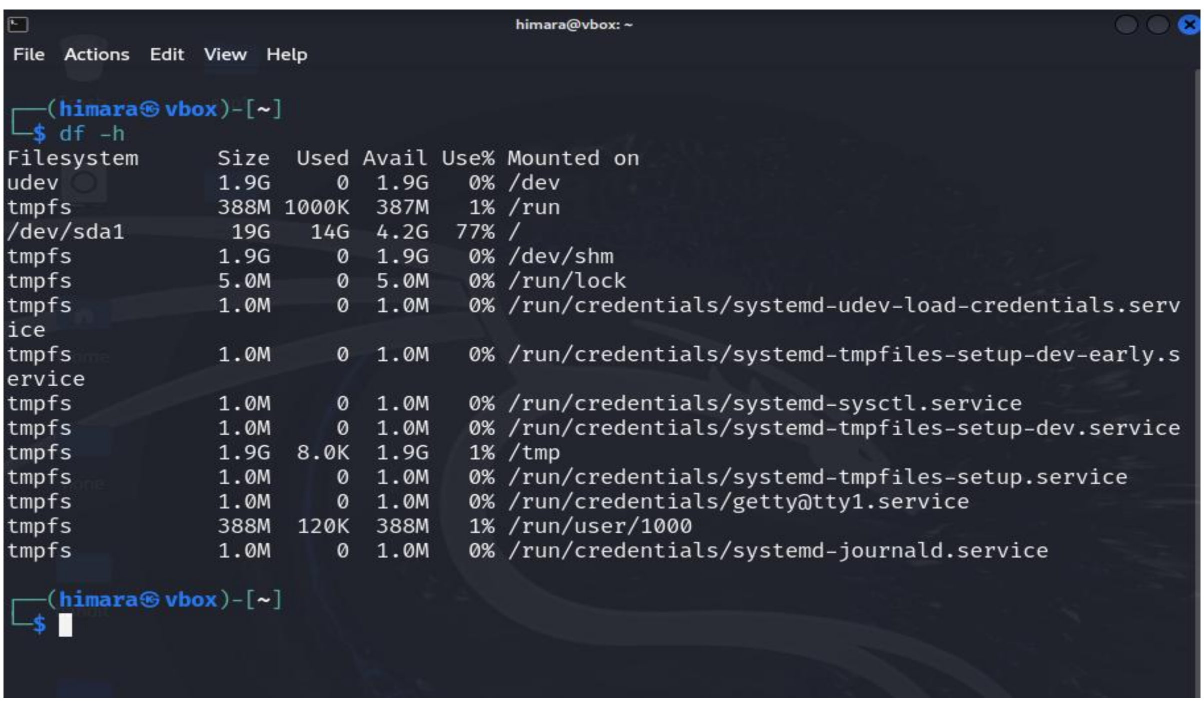
9. **cat – Concatenate and Display:** This is used for display or concatenate the content of a file. If it is an ASCII text file it will show everything inside that file.

10. **more:** This command shows a simple pager program that is used to display the text of a text file page by page. It allows one to read large files in a manageable fashion by scrolling page by page.

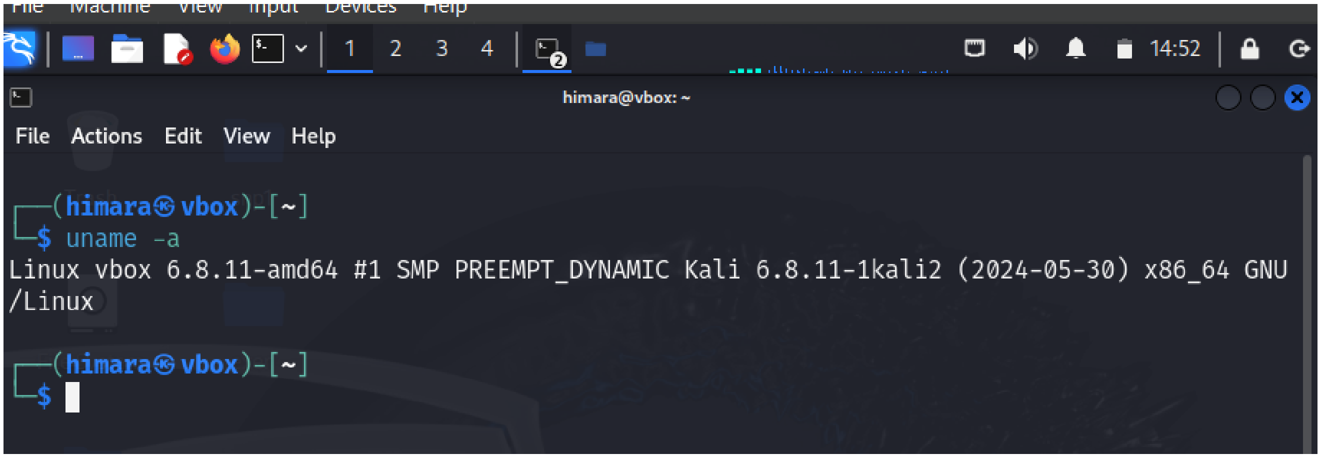


III. System Information and User Management

1. **df – Disk free:** Shows disk usage in human-readable format (GB/MB). This command is helps track how full your partitions are. We can used the "-h" flag the size information displayed in units of 1024 i. e. k, M. G and T. The default will be showing the sizes in Kilobytes if the "-h" flag is not added.



2. **uname -a:** This command displays system information like kernel version and architecture.

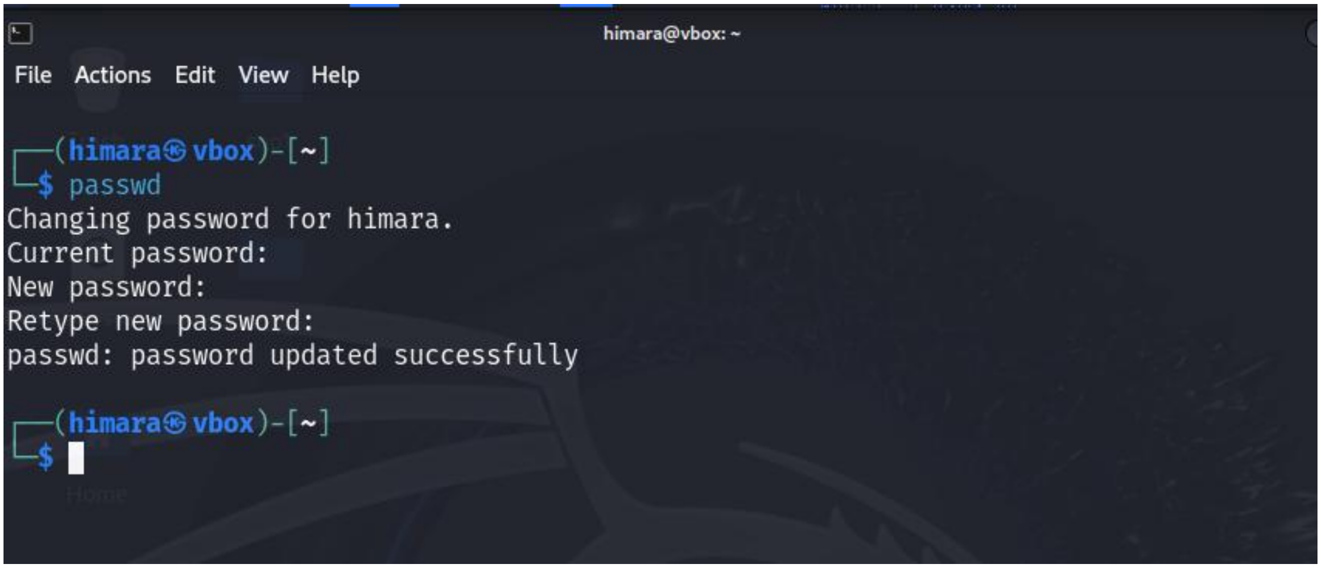


3. **free:** This command displays memory usage details of main memory (RAM) and swap memory. The free command will show information pertaining to total memory, used memory, free memory, shared memory, memory that is buffered, memory that is cached, and available memory. The default will be kibibytes and that is the same as 1024 bytes in amount.

4. **whoami – Print effective user name:** Print the user name associated with the current effective user ID.

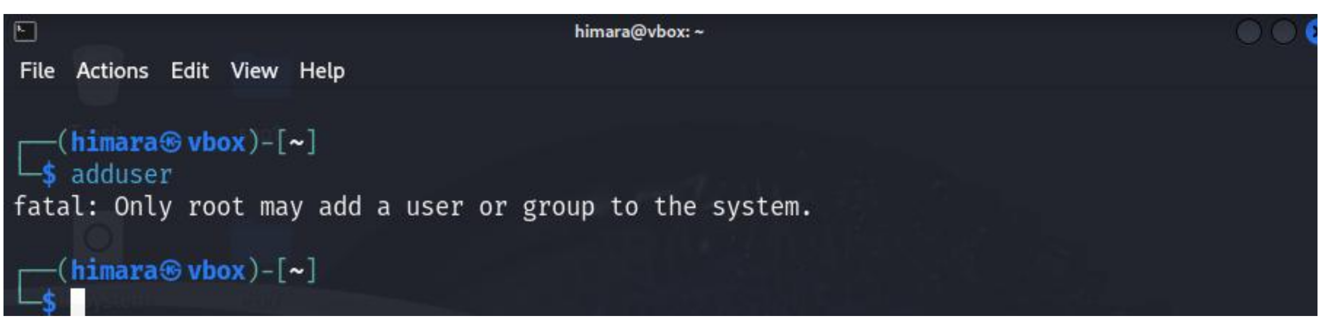
5. **hostname:** This command is use to display the system's DNS name, and to display or set its hostname or NIS domain name.

6. **passwd:** This command is use to change a user's password. Normal users can change their own and root can change any account. It asks the current password (if necessary), followed by twice the new password, and checks complexity and validity before actually changing it.



7. **who:** This command shows the lists all currently logged-in users and their session details.

8. **adduser:** Adds a new user. It also creates a home directory for the user.

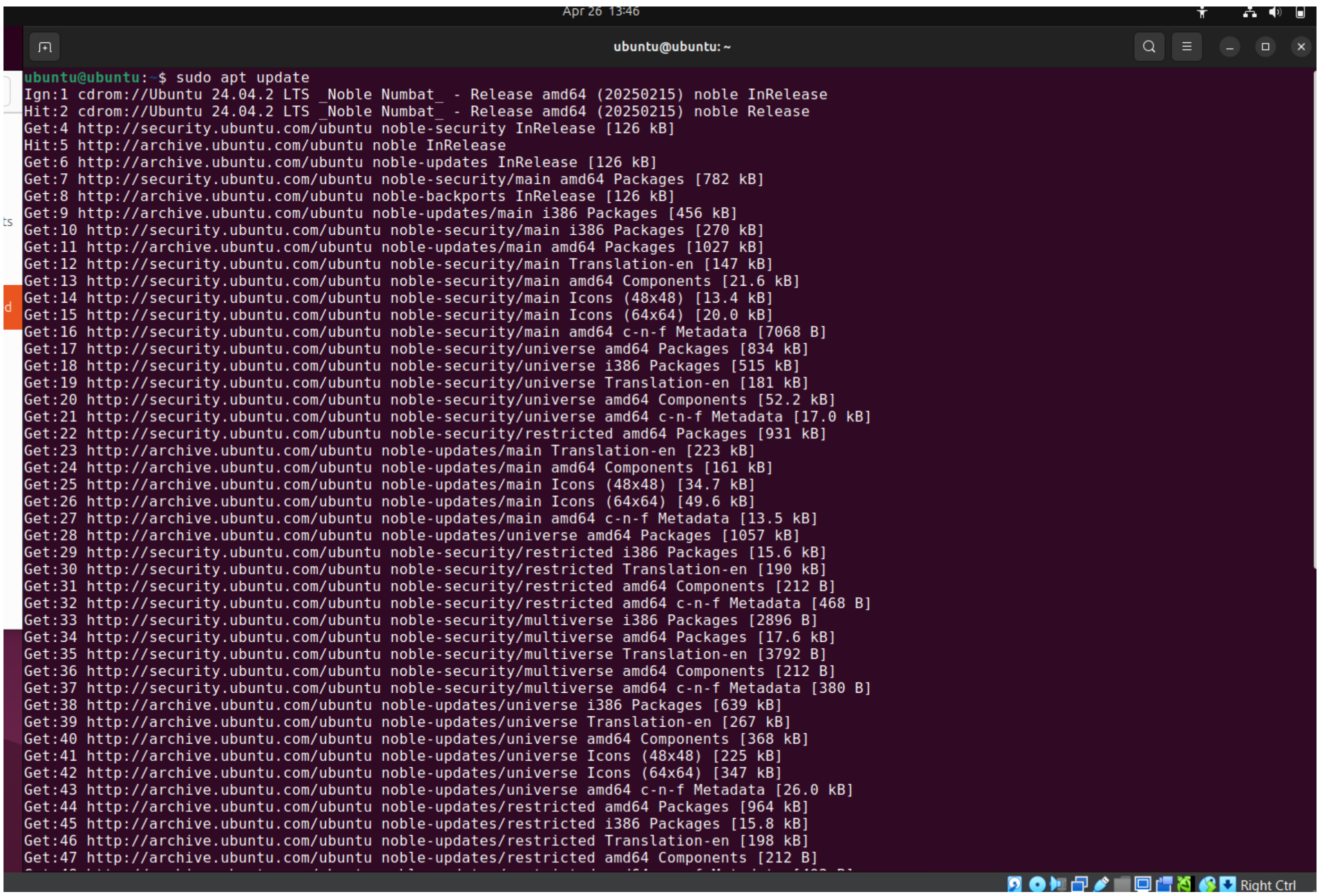


2. DHCP, DNS and NTP Services

1. DHCP (Dynamic Host Configuration Protocol)

1. Update the DHCP

First, execute 'apt update' to update the package index.



```
ubuntu@ubuntu:~$ sudo apt update
Ign:1 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble InRelease
Hit:2 cdrom://Ubuntu 24.04.2 LTS _Noble Numbat_ - Release amd64 (20250215) noble Release
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [782 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main i386 Packages [456 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [270 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1027 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [147 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main Icons (48x48) [13.4 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Icons (64x64) [20.0 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [7068 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [834 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [515 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [181 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [931 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [223 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [161 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (48x48) [34.7 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/main Icons (64x64) [49.6 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [13.5 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1057 kB]
Get:29 http://security.ubuntu.com/ubuntu noble-security/restricted i386 Packages [15.6 kB]
Get:30 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [190 kB]
Get:31 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:32 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [468 B]
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse i386 Packages [2896 B]
Get:34 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.6 kB]
Get:35 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]
Get:36 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:37 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-updates/universe i386 Packages [639 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [267 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [368 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (48x48) [225 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (64x64) [347 kB]
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Get:44 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [964 kB]
Get:45 http://archive.ubuntu.com/ubuntu noble-updates/restricted i386 Packages [15.8 kB]
Get:46 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [198 kB]
Get:47 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
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Get:40 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [368 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (48x48) [225 kB]
Get:42 http://archive.ubuntu.com/ubuntu noble-updates/universe Icons (64x64) [347 kB]
Get:43 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [26.0 kB]
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Get:59 http://archive.ubuntu.com/ubuntu noble-backports/main Icons (64x64) [11.2 kB]
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Get:61 http://archive.ubuntu.com/ubuntu noble-backports/universe i386 Packages [14.0 kB]
Get:62 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [27.1 kB]
Get:63 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [16.5 kB]
Get:64 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [16.3 kB]
Get:65 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (48x48) [20.4 kB]
Get:66 http://archive.ubuntu.com/ubuntu noble-backports/universe Icons (64x64) [28.7 kB]
Get:67 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1304 B]
Get:68 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:69 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 10.7 MB in 15s (698 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
250 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$
```
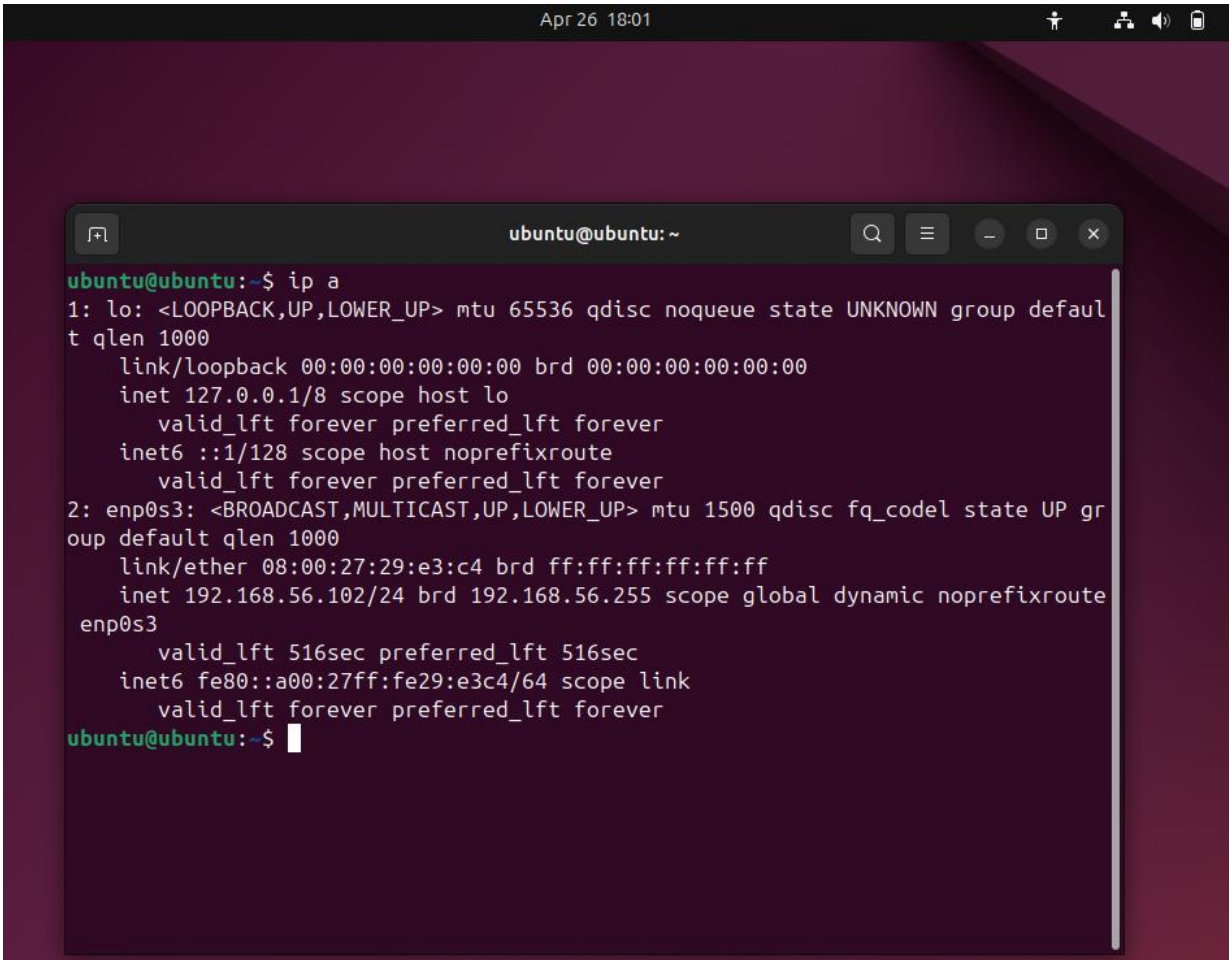

2.Install the DHCP server

Start by obtaining the **isc-dhcp-server** package. After that, execute the command **'apt install isc-dhcp-server'** to install and set it up on system.

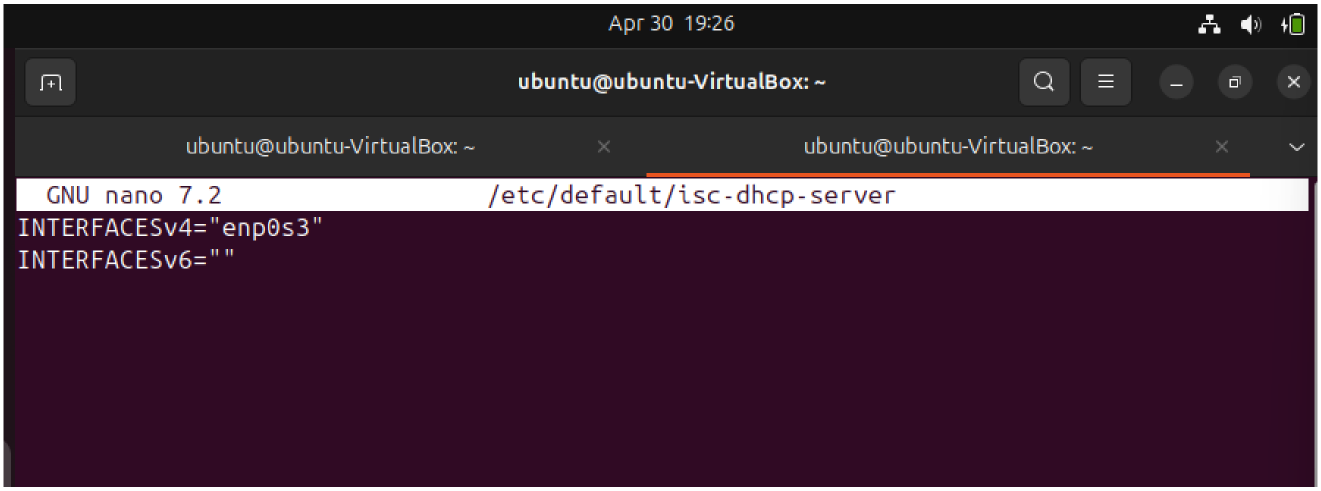
```
ubuntu@ubuntu:~$ sudo apt install isc-dhcp-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  isc-dhcp-common
Suggested packages:
  isc-dhcp-server-ldap policycoreutils
The following NEW packages will be installed:
  isc-dhcp-common isc-dhcp-server
0 upgraded, 2 newly installed, 0 to remove and 250 not upgraded.
Need to get 1281 kB of archives.
After this operation, 4281 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-server amd64 4.4.3-P1-4ubuntu2 [1236 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 isc-dhcp-common amd64 4.4.3-P1-4ubuntu2 [45.8 kB]
Fetched 1281 kB in 3s (401 kB/s)
Preconfiguring packages ...
Selecting previously unselected package isc-dhcp-server.
(Reading database ... 212032 files and directories currently installed.)
Preparing to unpack .../isc-dhcp-server_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Selecting previously unselected package isc-dhcp-common.
Preparing to unpack .../isc-dhcp-common_4.4.3-P1-4ubuntu2_amd64.deb ...
Unpacking isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Setting up isc-dhcp-server (4.4.3-P1-4ubuntu2) ...
Generating /etc/default/isc-dhcp-server...
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server.service → /usr/lib/systemd/system/isc-dhcp-server.service.
Created symlink /etc/systemd/system/multi-user.target.wants/isc-dhcp-server6.service → /usr/lib/systemd/system/isc-dhcp-server6.service.
Setting up isc-dhcp-common (4.4.3-P1-4ubuntu2) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu@ubuntu:~$
```

3. Configure the DHCP server

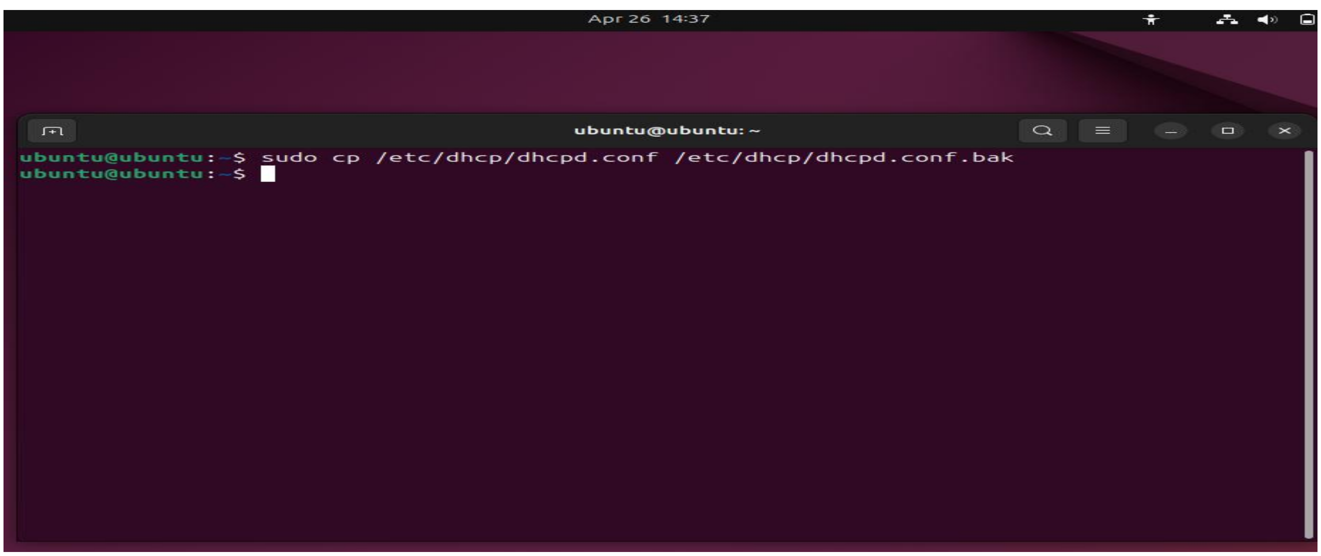
I need to get the IP address and network interface using the **‘ip a’** command.



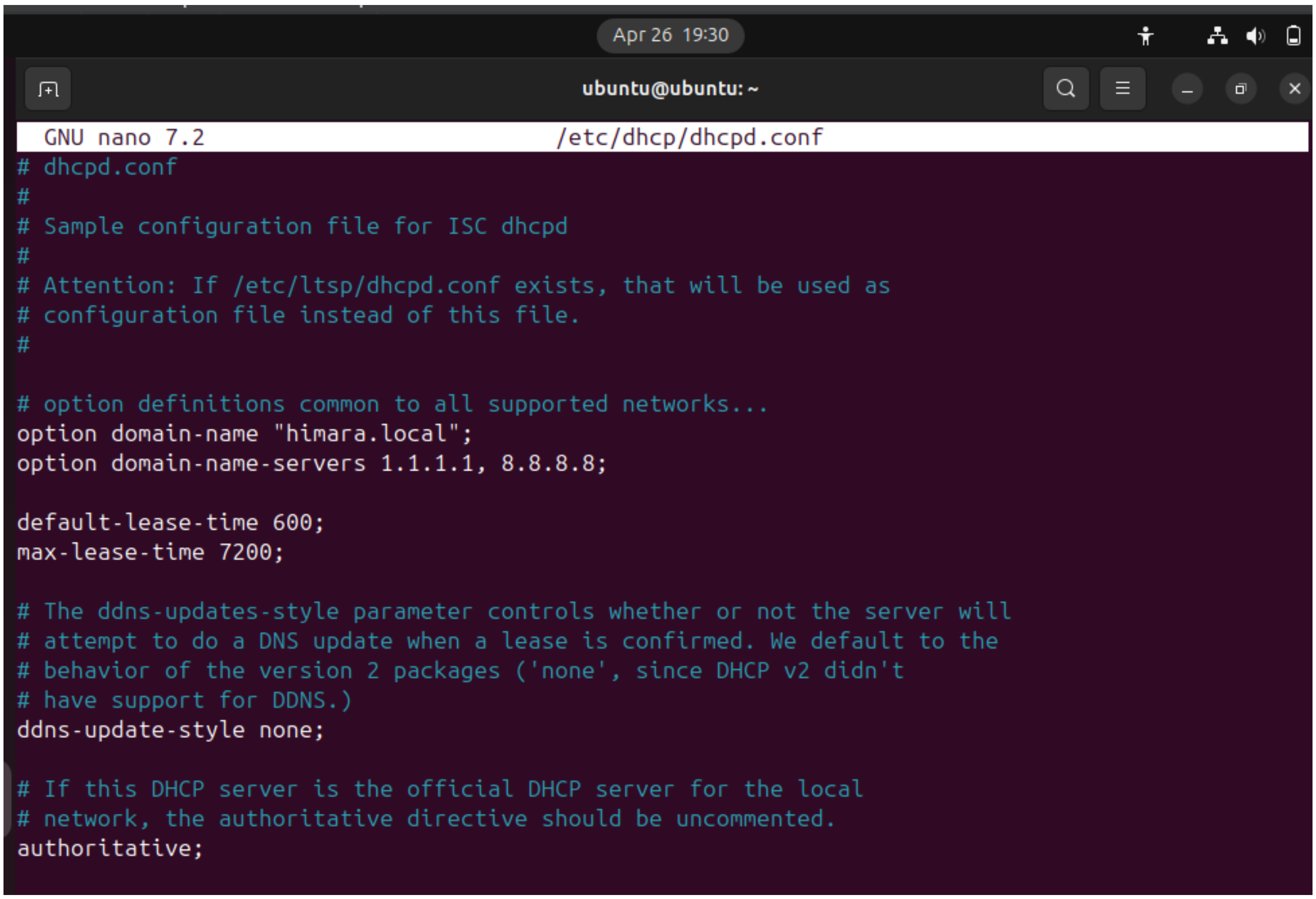
Edit the */etc/default/isc-dhcp-server* file to specify which interface the DHCP server should listen on and set the interface to “enp0s3”.

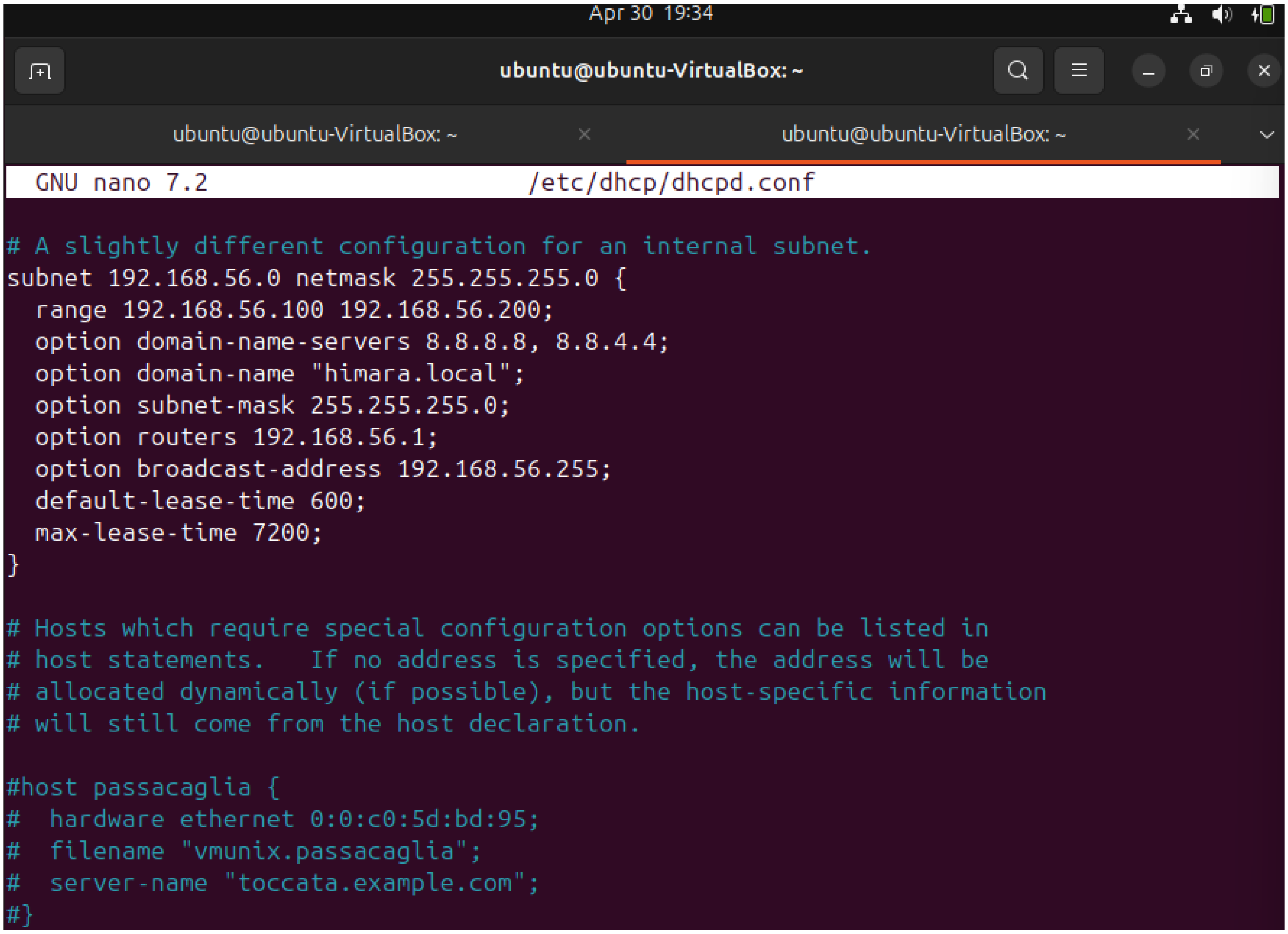


After installing the DHCP server the next step is to configure the DHCP server. - The configuration file for ISC DHCP is located at ***/etc/dhcp/dhcpd.conf***. Before making any changes, it’s wise to create a backup of the original file using ***cp*.**



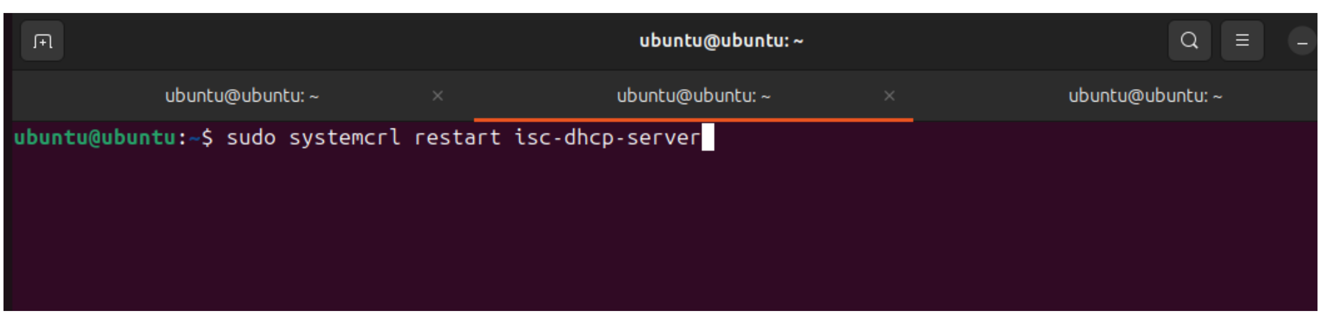
Next edit the /etc/dhcp/dhcpd.conf file to add or modify the configuration according to your network. Make the following modifications, then save the file and exit (ctrl+O press Enter and ctrl+X).





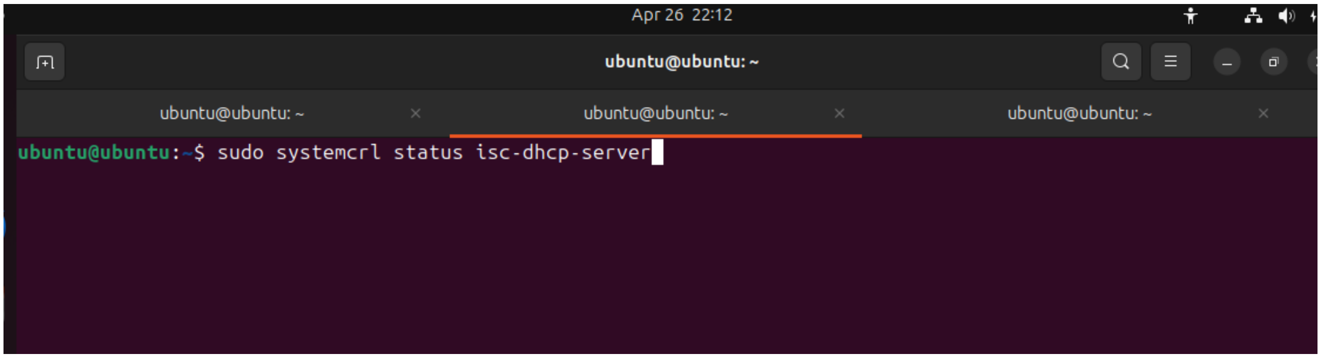
4. Restart the DHCP server

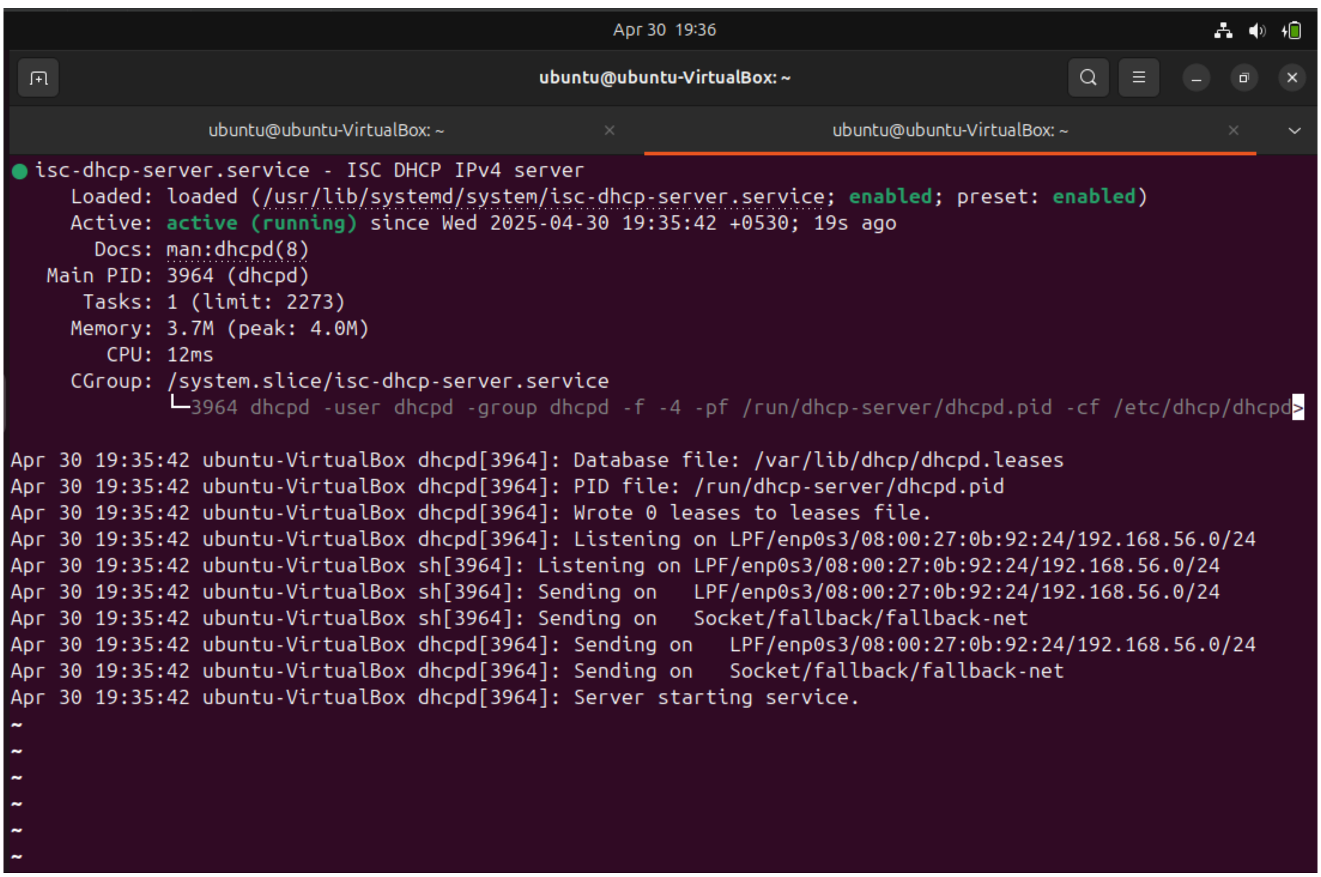
After finish the necessary configuration restart the DHCP server and save the changes.



5. Check the Status of the Service

The service ought to appear as active (running) on your list. Examine the configuration and log files for problems if there are any failures.

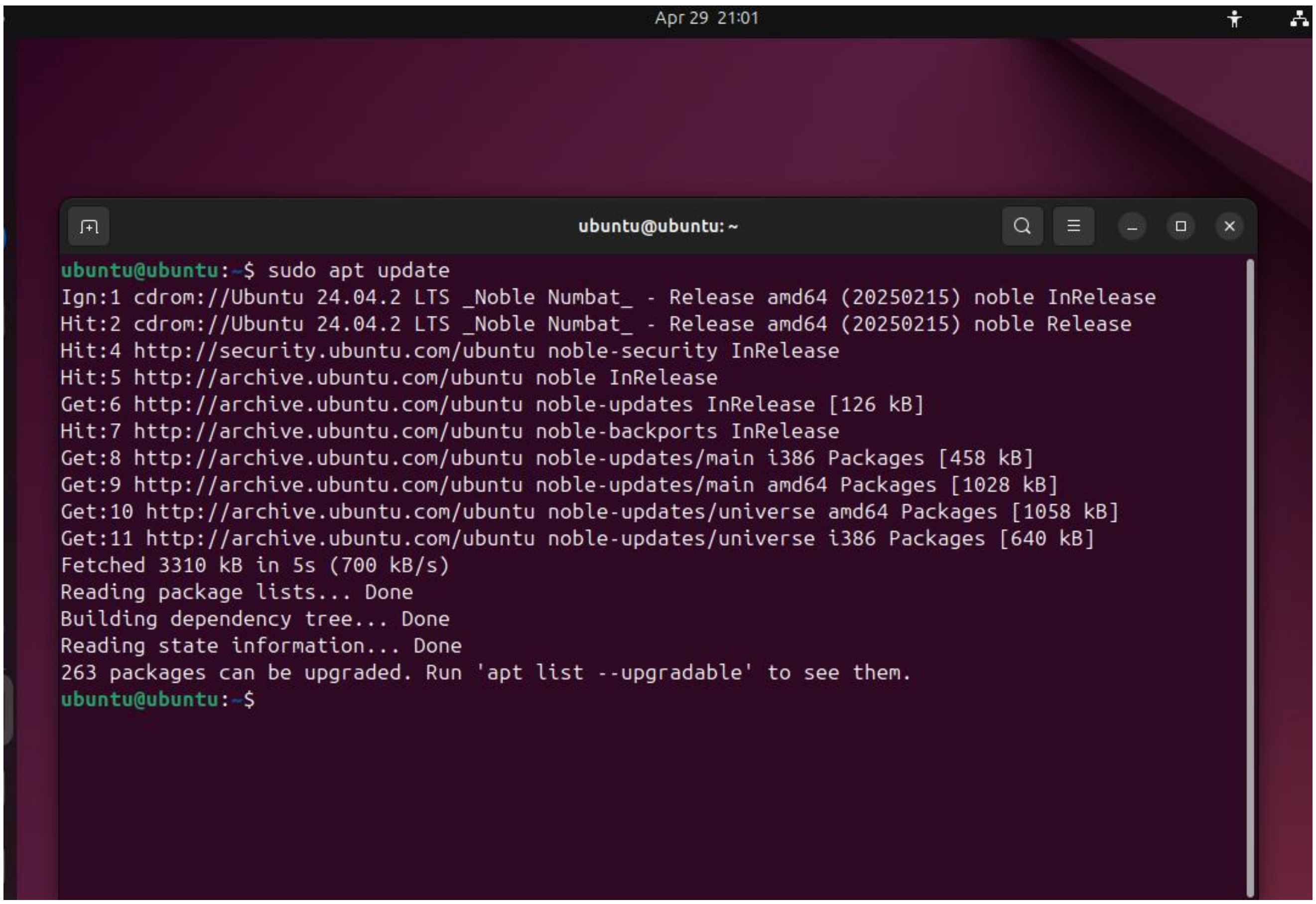




2. DNS (Domani Name Server)

Configuration of DNS server

1. **Update the Ubuntu system**
   This will update the system and refreshes your system's package list from the Ubuntu repositories.

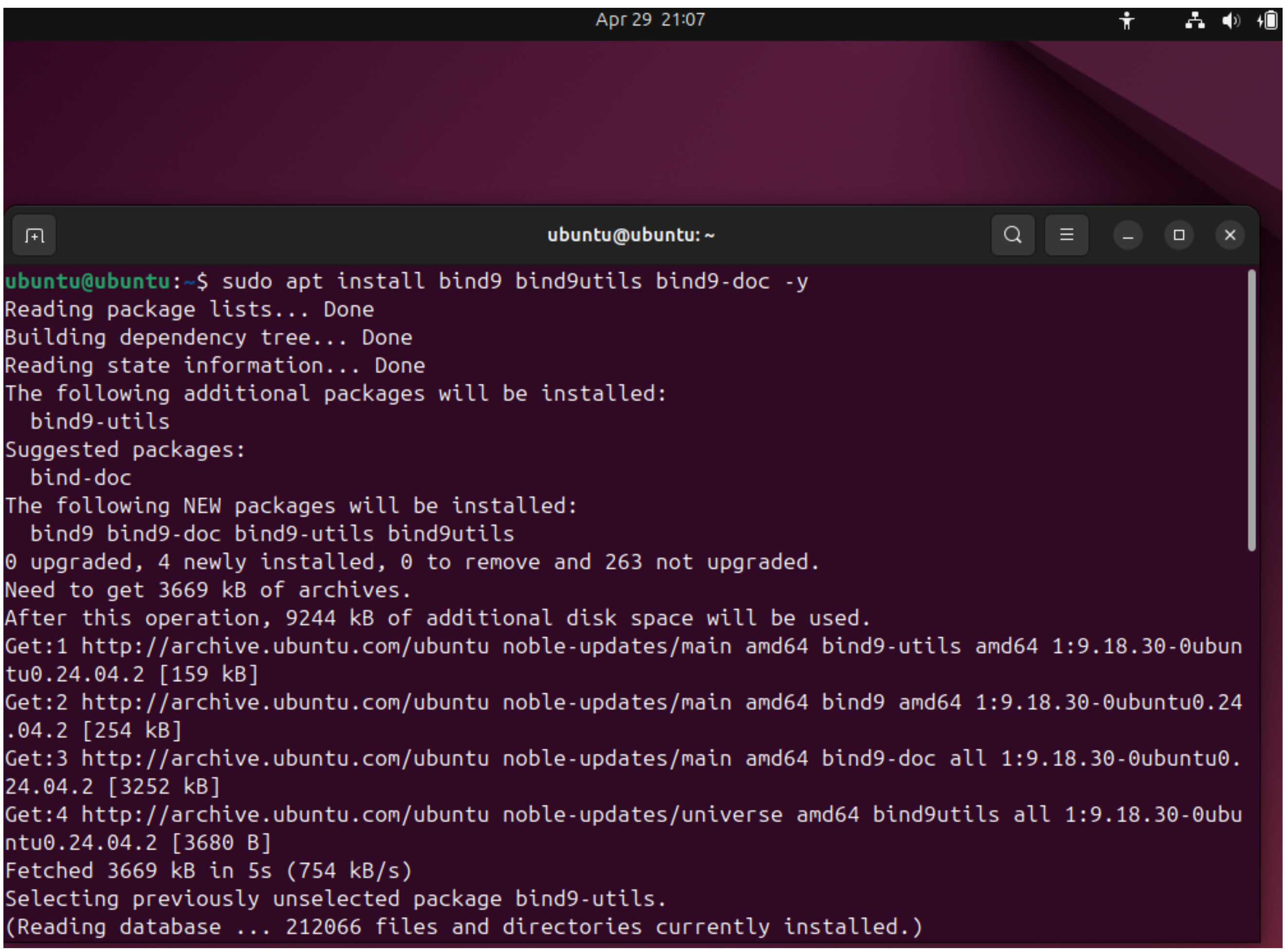


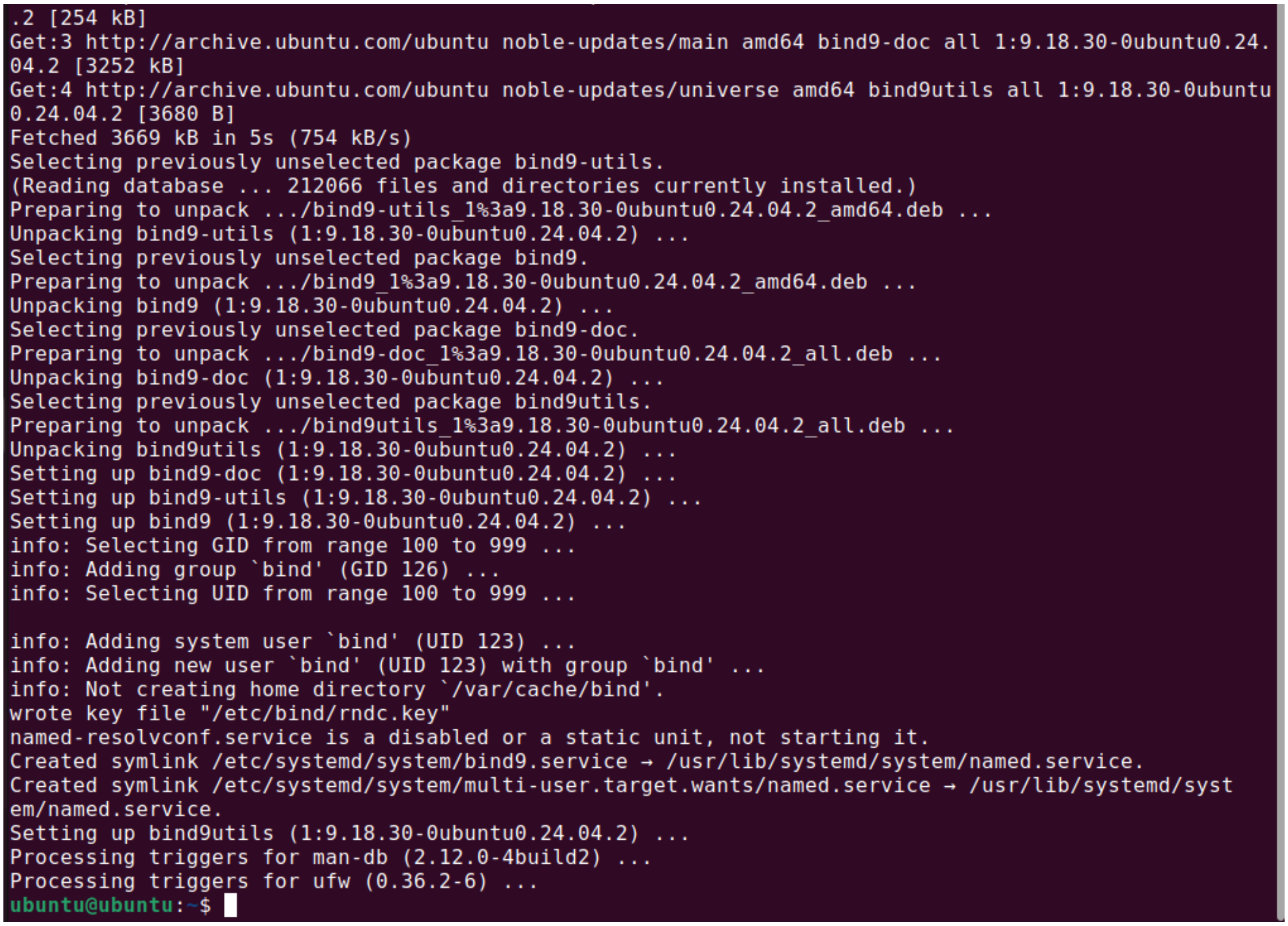
2. Install the BIND9

Now, get the package and run the command ‘sudo apt install bind9 bind9utils bind9-doc -y’ to set it up.

- **Bind (Berkeley Internet Name Domain)** is a widely used and highly reliable DNS server software.

- **bind9**: The DNS server software.
- **bind9utils**: Useful command-line utilities for BIND9.
- **bind9-doc**: Documentation for BIND9 (optional).

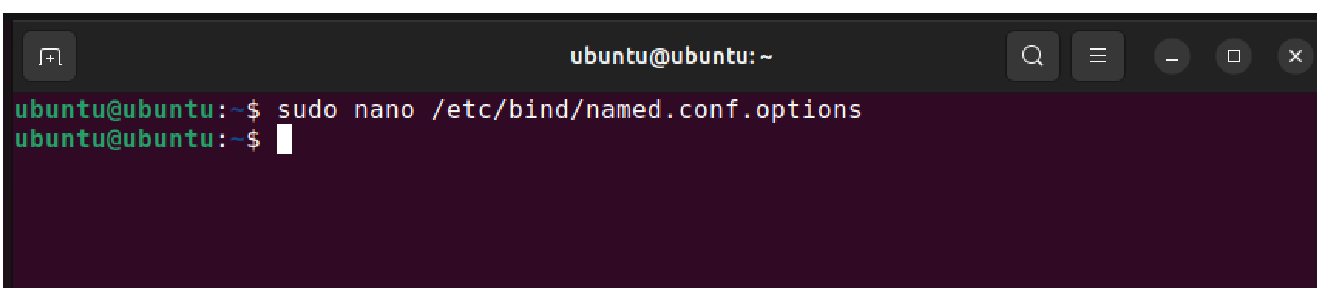




```
.2 [254 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-doc all 1:9.18.30-0ubuntu0.24.
04.2 [3252 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 bind9utils all 1:9.18.30-0ubuntu
0.24.04.2 [3680 B]
Fetched 3669 kB in 5s (754 kB/s)
Selecting previously unselected package bind9-utils.
(Reading database ... 212066 files and directories currently installed.)
Preparing to unpack .../bind9-utils_1%3a9.18.30-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9-utils (1:9.18.30-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1%3a9.18.30-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9 (1:9.18.30-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9-doc.
Preparing to unpack .../bind9-doc_1%3a9.18.30-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9-doc (1:9.18.30-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9utils.
Preparing to unpack .../bind9utils_1%3a9.18.30-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9utils (1:9.18.30-0ubuntu0.24.04.2) ...
Setting up bind9-doc (1:9.18.30-0ubuntu0.24.04.2) ...
Setting up bind9-utils (1:9.18.30-0ubuntu0.24.04.2) ...
Setting up bind9 (1:9.18.30-0ubuntu0.24.04.2) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `bind' (GID 126) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user `bind' (UID 123) ...
info: Adding new user `bind' (UID 123) with group `bind' ...
info: Not creating home directory `/var/cache/bind'.
wrote key file "/etc/bind/rndc.key"
named-resolvconf.service is a disabled or a static unit, not starting it.
Created symlink /etc/systemd/system/bind9.service → /usr/lib/systemd/system/named.service.
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/syst
em/named.service.
Setting up bind9utils (1:9.18.30-0ubuntu0.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
ubuntu@ubuntu:~$
```

3. Configure BIND9

- The named.conf file is BIND 9's primary configuration file.
- We can declare options we need for our configuration.

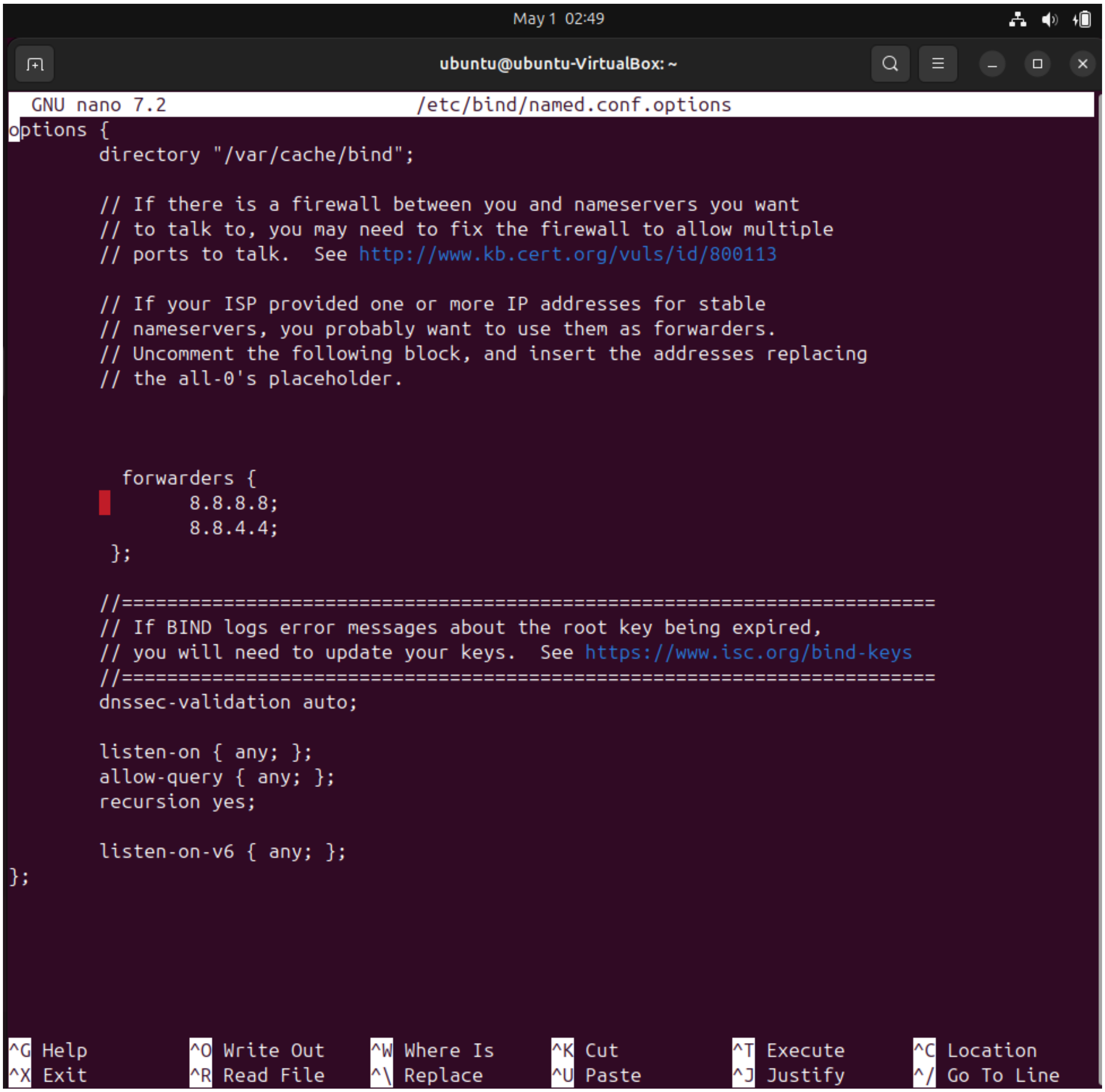



```
ubuntu@ubuntu:~$ sudo nano /etc/bind/named.conf.options
ubuntu@ubuntu:~$
```

- After that opent the file and do the following editors and the necessary codes should be uncommented and be modified.

  8.8.8.8: Google DNS

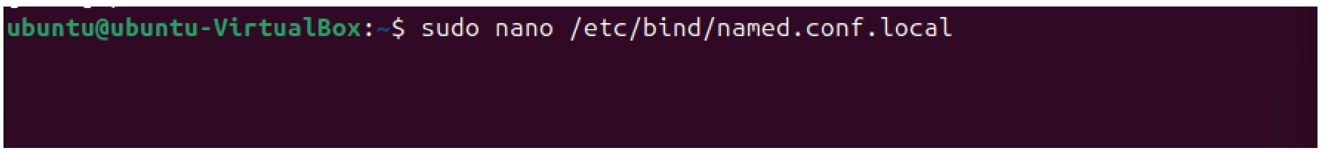
  8.8.4.4: Google DNS



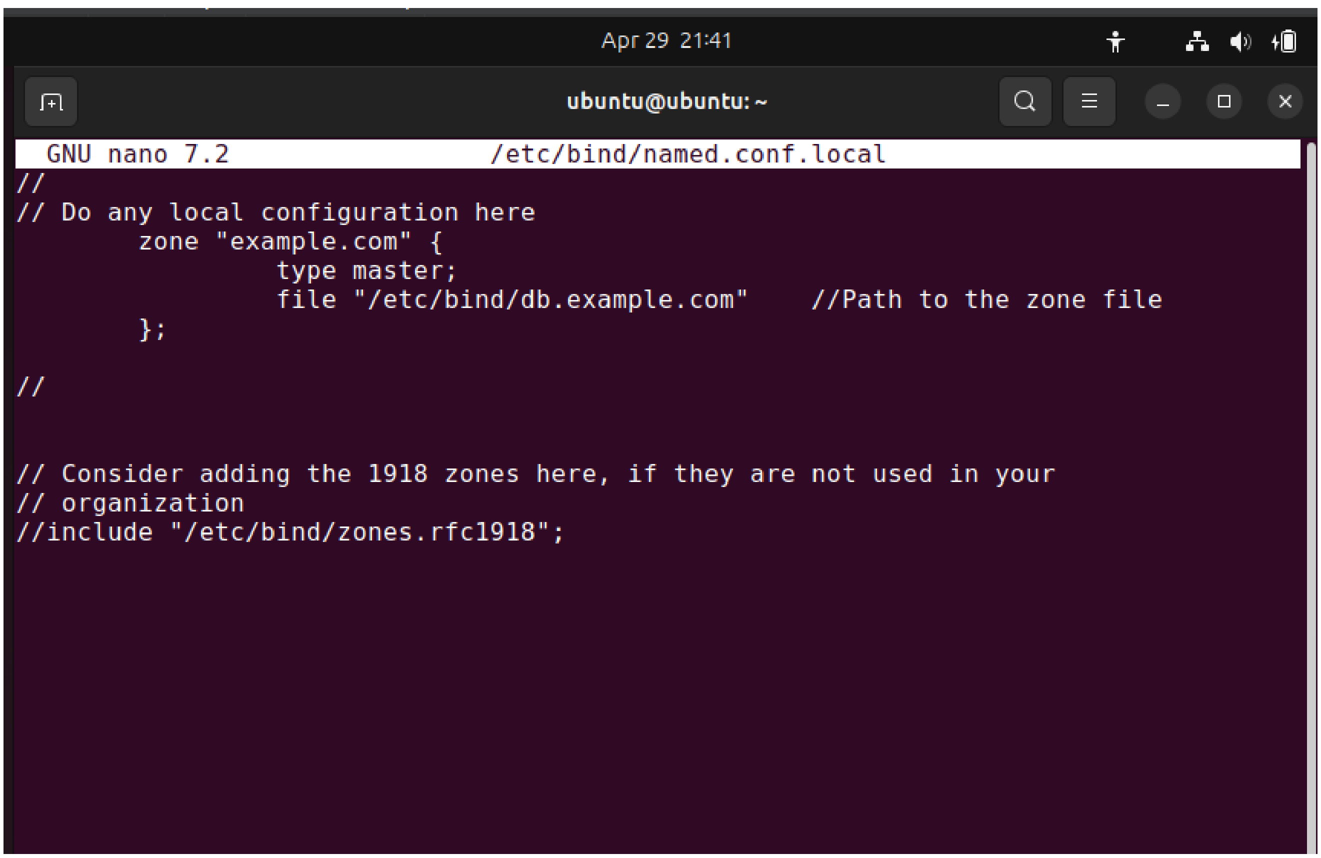
These are changed so that they send DNS requests that your server cannot resolve to other DNS servers.

4. Configure Zone Files

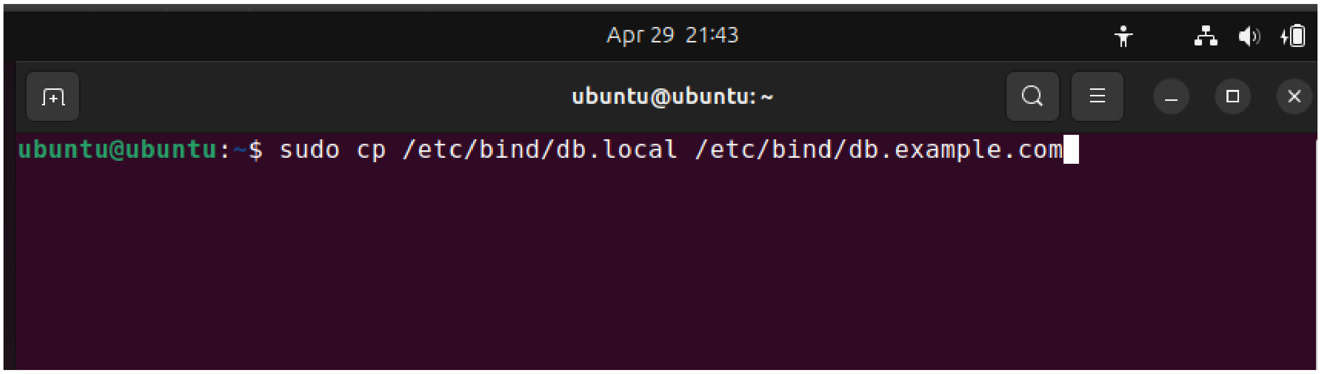
- To define your own DNS zones (for internal or external domains), you'll need to add the zone definition in the *named.conf.local* file.
- A **DNS zone** represents a portion of the DNS namespace and contains resource records that map domain names to IP addresses.

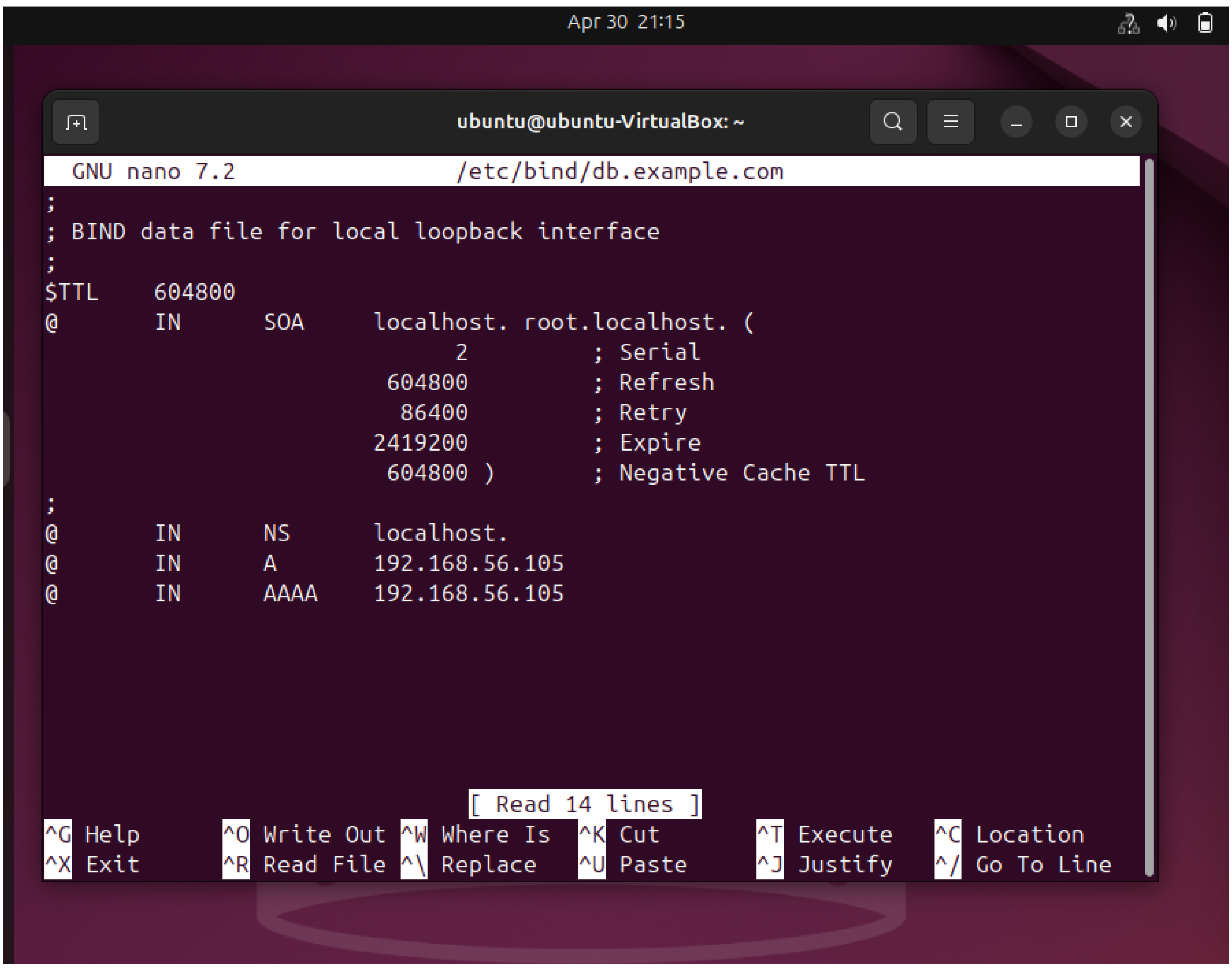


Edit the file following instructions to define the zone.



Next create a new zone file for your domain and open the newly created file and configure your DNS records.



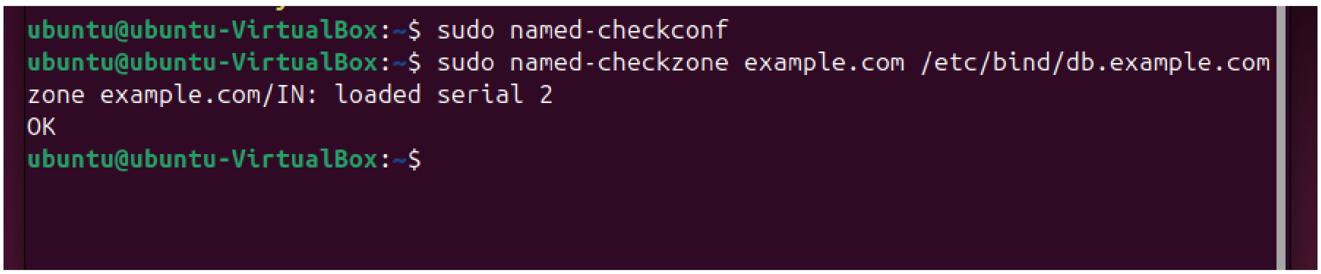


The ip address is should be changed to my IP address 192.168.56.105.

5. Check the configuration

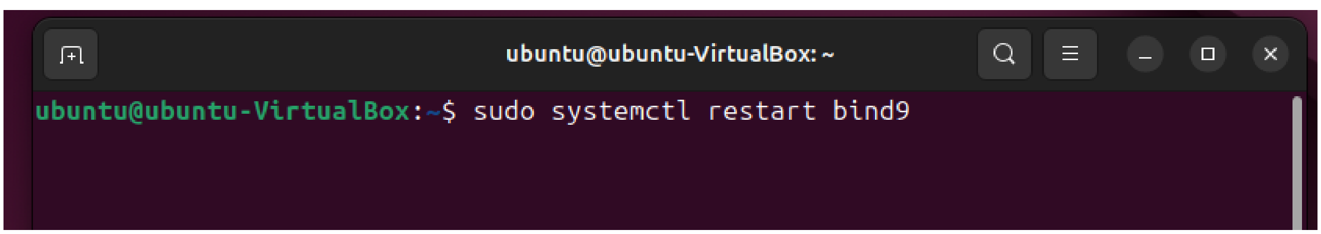
Check the syntax of BIND configuration and run the following command to check for errors in the configuration.

- **"sudo named-checkconf "** this command is used to check the zone file.
- **"sudo named-checkzone example.com /etc/bind/db.example.com"** this command is used to and verify that zone file are correctly formed.

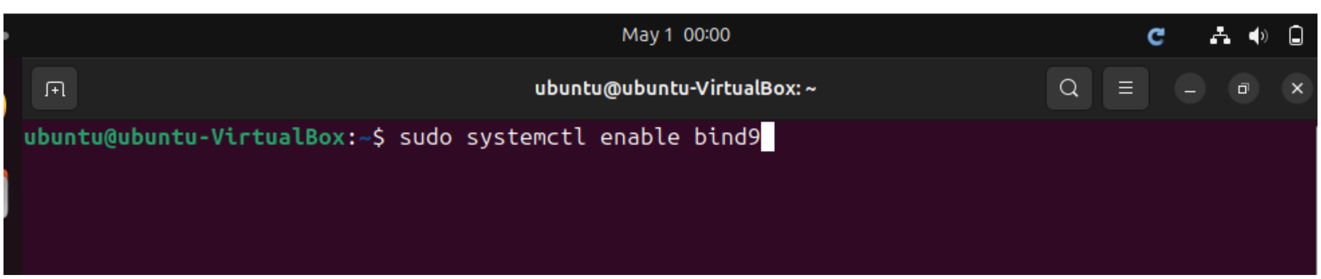


6. Start BIND

To apply the configuration changes, restart the DNS server.



7. Enable BIND9 to start on boot



Now use the **'sudo systemctl status bind9'** command and check the status of the server. OK It is (Active) running.

```
Apr 30 23:34
ubuntu@ubuntu-VirtualBox: ~
● named.service - BIND Domain Name Server
     Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
     Active: active (running) since Wed 2025-04-30 23:31:22 +0530; 33s ago
       Docs: man:named(8)
   Main PID: 27117 (named)
     Status: "running"
      Tasks: 8 (limit: 2273)
     Memory: 6.0M (peak: 6.3M)
        CPU: 60ms
     CGroup: /system.slice/named.service
             └─27117 /usr/sbin/named -f -u bind

Apr 30 23:31:22 ubuntu-VirtualBox named[27117]: zone localhost/IN: loaded serial 2
Apr 30 23:31:22 ubuntu-VirtualBox named[27117]: zone 0.in-addr.arpa/IN: loaded serial 1
Apr 30 23:31:22 ubuntu-VirtualBox named[27117]: zone example.com/IN: loaded serial 2
Apr 30 23:31:22 ubuntu-VirtualBox named[27117]: zone 127.in-addr.arpa/IN: loaded serial 1
Apr 30 23:31:22 ubuntu-VirtualBox named[27117]: all zones loaded
Apr 30 23:31:22 ubuntu-VirtualBox named[27117]: running
Apr 30 23:31:22 ubuntu-VirtualBox named[27117]: zone example.com/IN: sending notifies (serial 2)
Apr 30 23:31:22 ubuntu-VirtualBox systemd[1]: Started named.service - BIND Domain Name Server.
Apr 30 23:31:23 ubuntu-VirtualBox named[27117]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance timer complete)
Apr 30 23:31:23 ubuntu-VirtualBox named[27117]: managed-keys-zone: Key 38696 for zone . is now trusted (acceptance timer complete)
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
lines 1-22/22 (END)
```

8. Check BIND9 allow for the firewall - First

We have to configure the **bind9** service to be allowed through the firewall.

```
ubuntu@ubuntu-VirtualBox:~$ sudo ufw allow 53
Rules updated
Rules updated (v6)
ubuntu@ubuntu-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
53                         ALLOW       Anywhere
53 (v6)                    ALLOW       Anywhere (v6)

ubuntu@ubuntu-VirtualBox:~$
```

Now give a rule to BIND9 to go through the firewall.

```
ubuntu@ubuntu-VirtualBox:~$ sudo ufw allow bind9
Rule added
Rule added (v6)
ubuntu@ubuntu-VirtualBox:~$
```

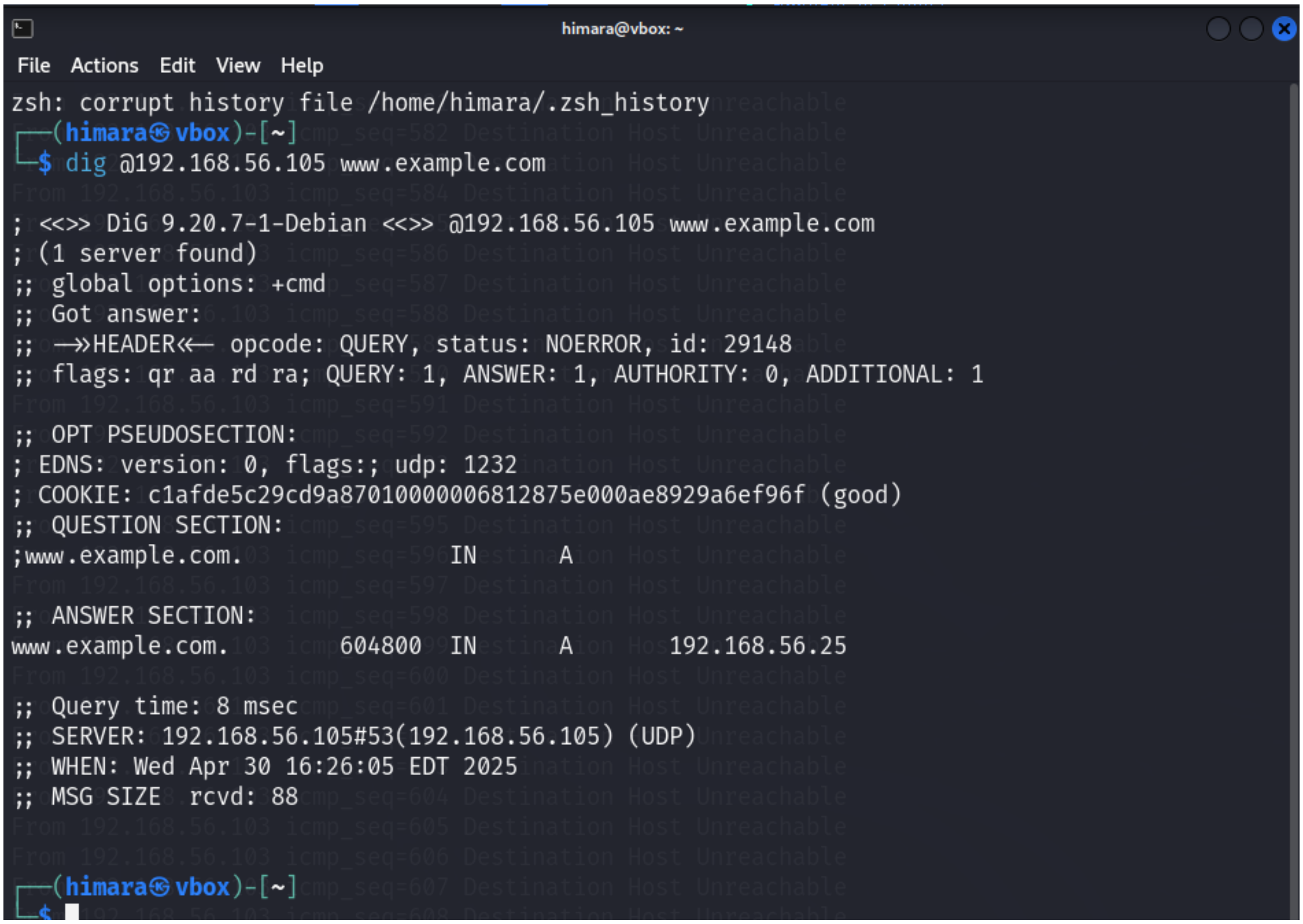
Now reload the ufw.

```
May 1  00:10
ubuntu@ubuntu-VirtualBox: ~
ubuntu@ubuntu-VirtualBox:~$ sudo ufw reload
Firewall reloaded
ubuntu@ubuntu-VirtualBox:~$
```

```
ubuntu@ubuntu-VirtualBox:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
53                         ALLOW       Anywhere
Bind9                      ALLOW       Anywhere
53 (v6)                    ALLOW       Anywhere (v6)
Bind9 (v6)                 ALLOW       Anywhere (v6)

ubuntu@ubuntu-VirtualBox:~$
```

9. Checking whether the DNS server works

This output shows the result of a DNS query for domain www.example.com that has been successfully resolved by the DNS server with IP address 192.168.56.105 and which has replied with IP address 192.168.56.25. The DNS query and response process was carried out without any errors



3. NTP (Network Time Protocol)

1. Update the Linux system

"sudo apt update" using this command you can update the your Linux system.

```
himara@vbox: ~                                   CPU usage: 0.0%
File  Actions  Edit  View  Help
zsh: corrupt history file /home/himara/.zsh_history
┌──(himara㉿vbox)-[~]
└─$ sudo apt update
[sudo] password for himara:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
2122 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

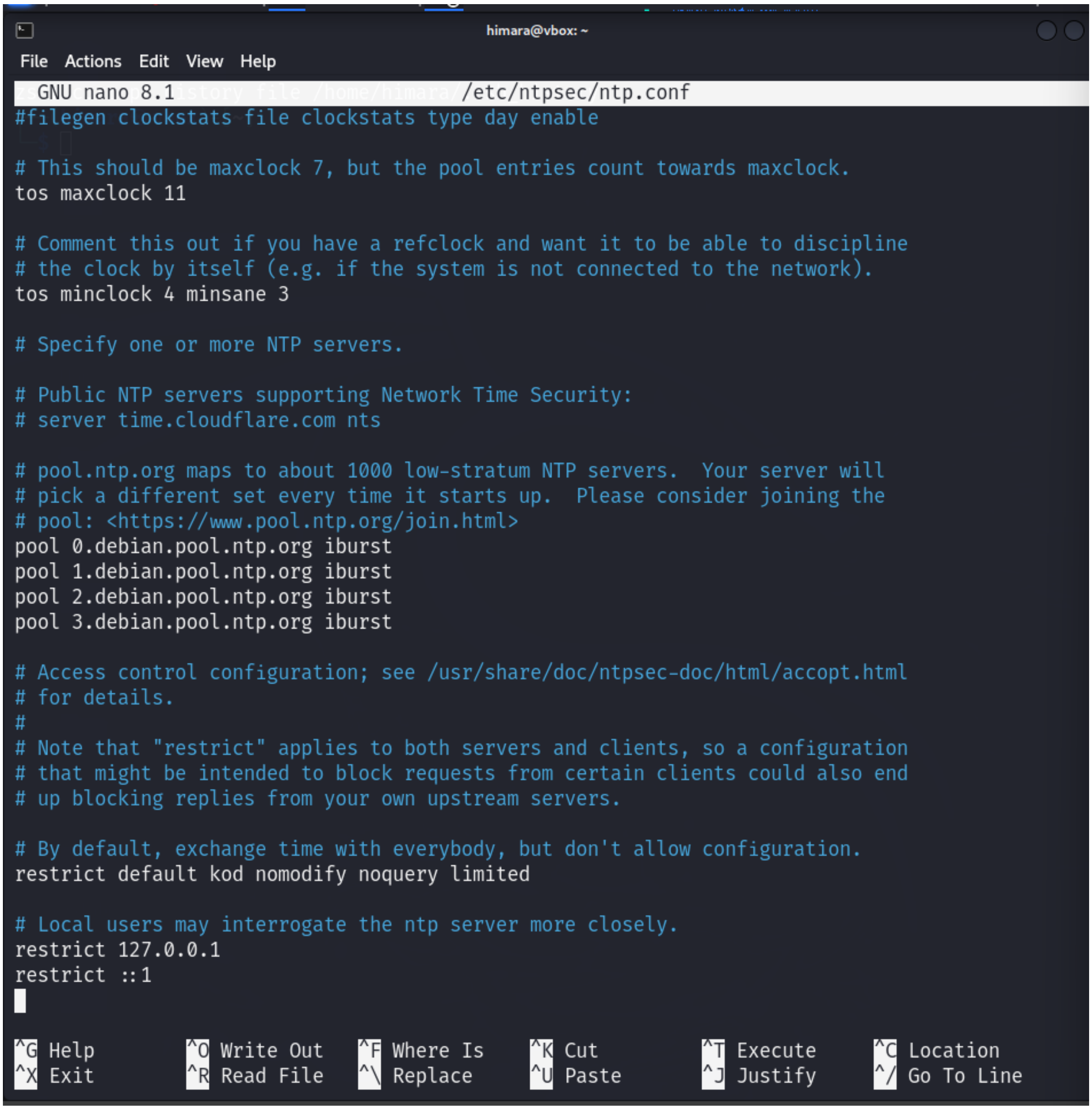
2. Install NTP

"sudo apt install ntp -y" using this command you can install the NTP package. The NTP package provides both the NTP daemon and the tools necessary to synchronize time.

```
himara@vbox: ~
File  Actions  Edit  View  Help
┌──(himara㉿vbox)-[~]
└─$ sudo apt install ntp -y

Installing:
  ntp

Installing dependencies:
  ntpsec  python3-ntp

Suggested packages:
  certbot  ntpsec-doc  ntpsec-ntpviz

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 2121
  Download size: 459 kB
  Space needed: 1364 kB / 1162 MB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 python3-ntp amd64 1.2.3+dfsg1-3 [90.3 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 ntpsec amd64 1.2.3+dfsg1-3 [345 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 ntp all 1:4.2.8p15+dfsg-2~1.2.3+dfsg1-3 [23.4 kB]
Fetched 459 kB in 1s (312 kB/s)
Selecting previously unselected package python3-ntp.
(Reading database ... 396626 files and directories currently installed.)
Preparing to unpack .../python3-ntp_1.2.3+dfsg1-3_amd64.deb ...
Unpacking python3-ntp (1.2.3+dfsg1-3) ...
Selecting previously unselected package ntpsec.
Preparing to unpack .../ntpsec_1.2.3+dfsg1-3_amd64.deb ...
Unpacking ntpsec (1.2.3+dfsg1-3) ...
Selecting previously unselected package ntp.
Preparing to unpack .../ntp_1%3a4.2.8p15+dfsg-2~1.2.3+dfsg1-3_all.deb ...
Unpacking ntp (1:4.2.8p15+dfsg-2~1.2.3+dfsg1-3) ...
Setting up python3-ntp (1.2.3+dfsg1-3) ...
Setting up ntpsec (1.2.3+dfsg1-3) ...
update-rc.d: We have no instructions for the ntpsec init script.
update-rc.d: It looks like a network service, we disable it.
Created symlink '/etc/systemd/system/timers.target.wants/ntpsec-rotate-stats.timer' → '/usr/lib/systemd/system/nt
psec-rotate-stats.timer'.
Created symlink '/etc/systemd/system/network-pre.target.wants/ntpsec-systemd-netif.path' → '/usr/lib/systemd/syst
em/ntpsec-systemd-netif.path'.
ntpsec.service is a disabled or a static unit, not starting it.
```

3. Configure NTP server

Use thi command **"sudo nano /etc/ntpsec/ntp.conf "** you can get a /etc/ntpsec/ntp.conf file and you can edit the main configuration file for NTP. After open the file we can replace or modify the lines with the NTP server you want to use.



```
himara@vbox: ~
File  Actions  Edit  View  Help
GNU nano 8.1                    /etc/ntpsec/ntp.conf
#filegen clockstats file clockstats type day enable

# This should be maxclock 7, but the pool entries count towards maxclock.
tos maxclock 11

# Comment this out if you have a refclock and want it to be able to discipline
# the clock by itself (e.g. if the system is not connected to the network).
tos minclock 4 minsane 3

# Specify one or more NTP servers.

# Public NTP servers supporting Network Time Security:
# server time.cloudflare.com nts

# pool.ntp.org maps to about 1000 low-stratum NTP servers.  Your server will
# pick a different set every time it starts up.  Please consider joining the
# pool: <https://www.pool.ntp.org/join.html>
pool 0.debian.pool.ntp.org iburst
pool 1.debian.pool.ntp.org iburst
pool 2.debian.pool.ntp.org iburst
pool 3.debian.pool.ntp.org iburst

# Access control configuration; see /usr/share/doc/ntpsec-doc/html/accopt.html
# for details.
#
# Note that "restrict" applies to both servers and clients, so a configuration
# that might be intended to block requests from certain clients could also end
# up blocking replies from your own upstream servers.

# By default, exchange time with everybody, but don't allow configuration.
restrict default kod nomodify noquery limited

# Local users may interrogate the ntp server more closely.
restrict 127.0.0.1
restrict ::1

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

4. Star and enable NTP

“sudo systemctl start ntpsec” Using this command now you can start and enable the NTP.

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl start ntpsec

┌──(himara㉿vbox)-[~]
└─$ sudo systemctl enable ntpsec
Synchronizing state of ntpsec.service with SysV service script with /usr/lib/systemd/systemd-s
ysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ntpsec
Created symlink '/etc/systemd/system/ntp.service' → '/usr/lib/systemd/system/ntpsec.service'.
Created symlink '/etc/systemd/system/ntpd.service' → '/usr/lib/systemd/system/ntpsec.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ntpsec.service' → '/usr/lib/syste
md/system/ntpsec.service'.

┌──(himara㉿vbox)-[~]
└─$ 
```

5. View the status of the NTP

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl status ntpsec
● ntpsec.service - Network Time Service
     Loaded: loaded (/usr/lib/systemd/system/ntpsec.service; enabled; preset: disabled)
     Active: active (running) since Thu 2025-05-01 16:10:49 EDT; 1min 16s ago
 Invocation: 4592f6bf6d3747e3ac1175ee97326928
       Docs: man:ntpd(8)
   Main PID: 7643 (ntpd)
      Tasks: 1 (limit: 4548)
     Memory: 13M (peak: 13.5M)
        CPU: 97ms
     CGroup: /system.slice/ntpsec.service
             └─7643 /usr/sbin/ntpd -p /run/ntpd.pid -c /etc/ntpsec/ntp.conf -g -N -u ntpsec:ntpsec

May 01 16:10:52 vbox ntpd[7643]: DNS: Pool taking: 2606:4700:f1::1
May 01 16:10:52 vbox ntpd[7643]: DNS: dns_take_status: 2.debian.pool.ntp.org⇒good, 8
May 01 16:10:53 vbox ntpd[7643]: DNS: dns_probe: 3.debian.pool.ntp.org, cast_flags:8, flags:101
May 01 16:10:53 vbox ntpd[7643]: DNS: dns_check: processing 3.debian.pool.ntp.org, 8, 101
May 01 16:10:53 vbox ntpd[7643]: DNS: Pool skipping: 222.165.180.134
May 01 16:10:53 vbox ntpd[7643]: DNS: Pool skipping: 162.159.200.123
May 01 16:10:53 vbox ntpd[7643]: DNS: Pool skipping: 162.159.200.1
May 01 16:10:53 vbox ntpd[7643]: DNS: dns_take_status: 3.debian.pool.ntp.org⇒good, 8
May 01 16:10:58 vbox ntpd[7643]: CLOCK: time stepped by 1.534429
May 01 16:10:58 vbox ntpd[7643]: INIT: MRU 10922 entries, 13 hash bits, 65536 bytes

┌──(himara㉿vbox)-[~]
└─$ 
```

6. Verify Time Synchronization

In Linux, the ntpq -p command is used to display a list of NTP peers that the local machine is synced with and to query Network Time Protocol (NTP) servers for their current status.

```
┌──(himara㉿vbox)-[~]
└─$ ntpq -p
     remote                                   refid      st t when poll reach   delay   offset  jitter
==============================================================================================================
 0.debian.pool.ntp.org                    .POOL.          16 p    -  256    0   0.0000   0.0000   0.0000
 1.debian.pool.ntp.org                    .POOL.          16 p    -  256    0   0.0000   0.0000   0.0000
 2.debian.pool.ntp.org                    .POOL.          16 p    -  256    0   0.0000   0.0000   0.0000
 3.debian.pool.ntp.org                    .POOL.          16 p    -  256    0   0.0000   0.0000   0.0000
+time.cloudflare.com                      10.154.8.7       3 u    5   64    7  67.1583   3.9268   8.6179
*ntp.sltidc.lk                            216.239.35.4     2 u    5   64    7  21.4303  -3.1807  99.4783
+time.cloudflare.com                      10.229.8.116     3 u    5   64    7  52.2257  -3.5106   5.8641
+time.cloudflare.com                      10.228.8.4       3 u    -   64   17  66.9259   4.8640 108.0435
+time.cloudflare.com                      10.156.9.195     3 u   63   64    7  66.2598   4.6179   6.5976

┌──(himara㉿vbox)-[~]
└─$ 
```

The output of timedatectl status shows that the system's local time is Eastern Daylight Time (EDT) and that the system is synchronized with NTP. The system is using UTC as the hardware clock (RTC), and the time zone is correctly set to America/New_York.

```
┌──(himara㉿vbox)-[~]
└─$ timedatectl status
               Local time: Thu 2025-05-01 16:16:45 EDT
           Universal time: Thu 2025-05-01 20:16:45 UTC
                 RTC time: Thu 2025-05-01 20:16:43
                Time zone: America/New_York (EDT, -0400)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no

┌──(himara㉿vbox)-[~]
└─$ 
```

IV. Need and use of the network services in administration

a) DHCP

- **Need:** DHCP is needed in order to have devices automatically configured in a network. Without DHCP, each device (computers, phones, printers) would have to be statically configured with an IP. Not only is this a laborious process, but it is also prone to errors. DHCP allows it to be simplified to such a degree so that it reduces administrative work and avoids conflicts when different devices would have the same IP.

- **Use:** They make devices part of a network and are configured with an IP address, subnet mask, etc., by automatically assigning them. DHCP ensures in a large network that each device will have a unique IP address without manual configuration by an administrator.

b) DNS

- **Need:** DNS is the internet "phonebook." It maps human-readable addresses (i.e., www.example.com) to computer-readable IP addresses so other devices can locate each other on a network. Without DNS, people would have to remember and type numeric IP addresses to access a website, making it impossible.

- **Use:** DNS facilitates internet browsing for users. DNS servers convert an entered website address in a browser into a relevant IP server address on which the site is hosted. DNS is involved in services for an internal network as well as browsing on the external internet.

c) NTP

- **Need:** Accurate timekeeping is crucial to a great deal of network functionality, including such things as event logging, data integrity and synchronizing equipment. Without synchronizing equipment to a universal time, events like automated backups, data transfer or security logs are undependable and in disarray.

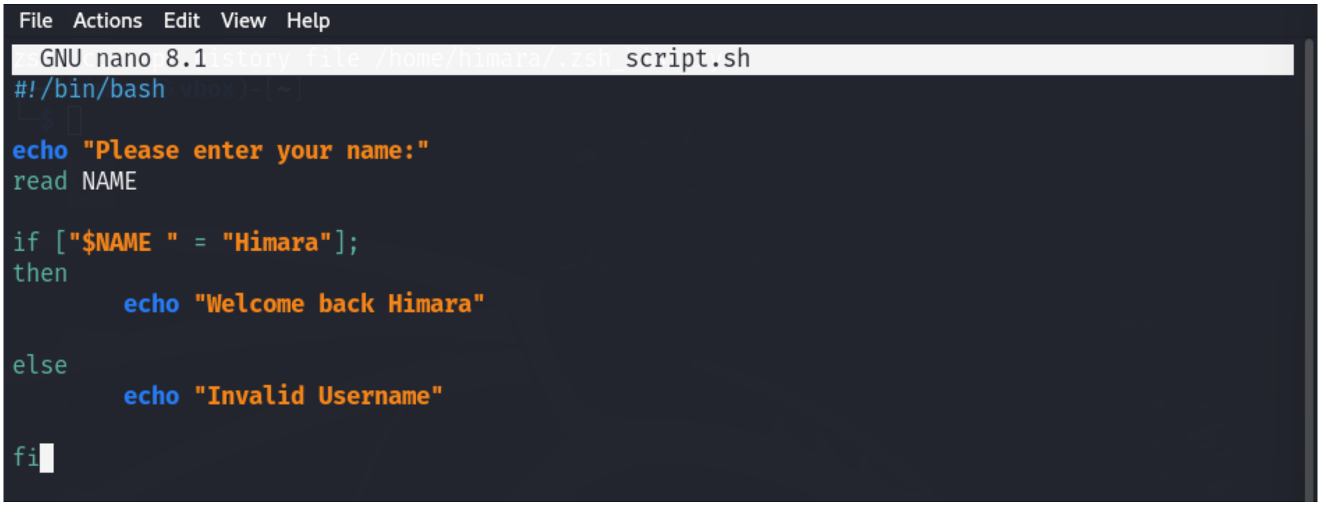
- **Use:** NTP synchronizes all devices in a network simultaneously without errors and discrepancies. NTP synchronizes computer clocks to an extremely precise time source such as an atomic clock or GPS so that devices will be accurate.

3. Security and other servers

1. Shell Scripting

Shell Scripting Basics:

• If-else statement


File Actions Edit View Help
GNU nano 8.1 script.sh
#!/bin/bash
echo "Please enter your name:"
read NAME
if ["$NAME " = "Himara"];
then
echo "Welcome back Himara"
else
echo "Invalid Username"
fi

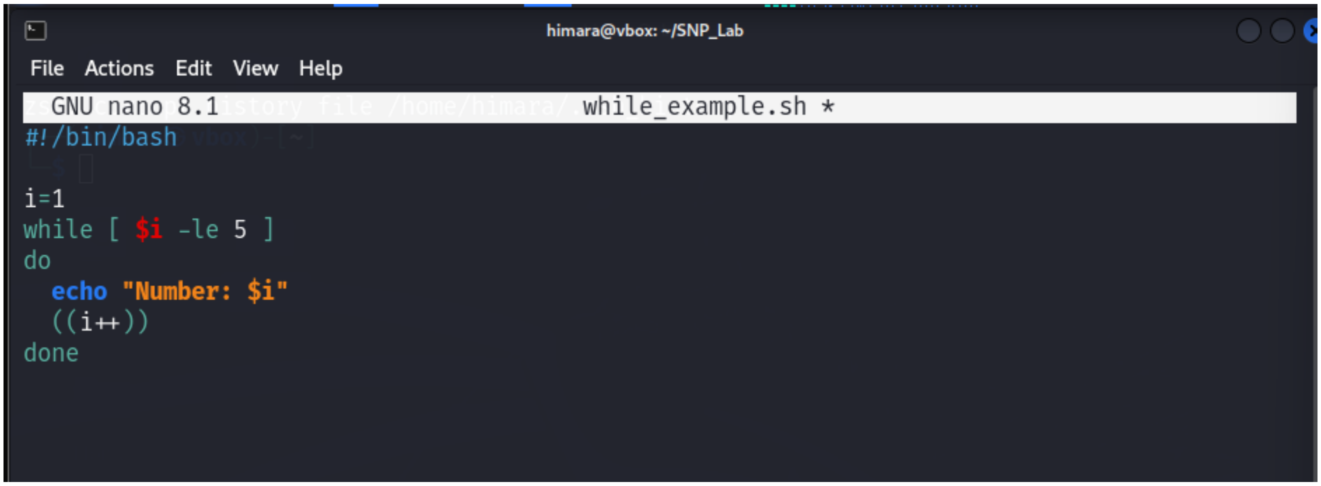
• For loop


himara@vbox: ~/SNP_Lab
File Actions Edit View Help
zsh: corrupt history file /home/himara/.zsh_history
(himara㉿vbox)-[~/SNP_Lab]
$ nano loop_example.sh
(himara㉿vbox)-[~/SNP_Lab]
$ chmod +x loop_example.sh
(himara㉿vbox)-[~/SNP_Lab]
$ ./loop_example.sh
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
(himara㉿vbox)-[~/SNP_Lab]
$

```
himara@vbox: ~/SNP_Lab
File  Actions  Edit  View  Help
  GNU nano 8.1                      function_example.sh *
#!/bin/bash

print_numbers() {
  for i in {1..5}
  do
     echo "Number: $i"
  done
}

print_numbers
```

• While loop

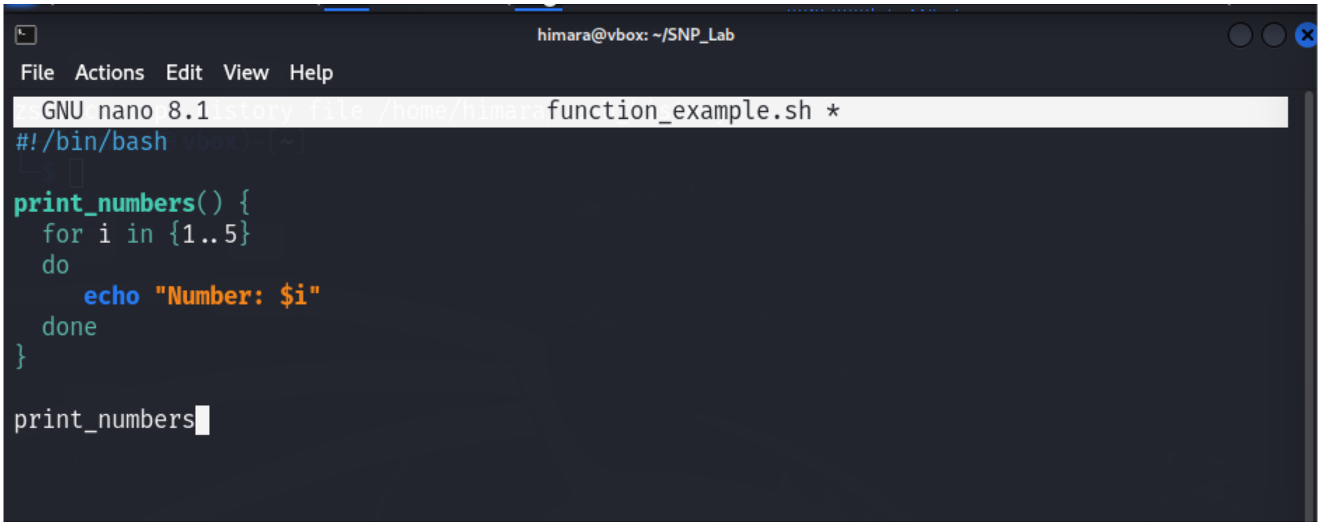
```
┌──(himara㉿vbox)-[~/SNP_Lab]
└─$ nano while_example.sh

┌──(himara㉿vbox)-[~/SNP_Lab]
└─$ chmod +x while_example.sh

┌──(himara㉿vbox)-[~/SNP_Lab]
└─$ ./while_example.sh
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5

┌──(himara㉿vbox)-[~/SNP_Lab]
└─$ 
```



himara@vbox: ~/SNP_Lab
File Actions Edit View Help
GNU nano 8.1 while_example.sh *
#!/bin/bash
i=1
while [$i -le 5]
do
echo "Number: $i"
((i++))
done

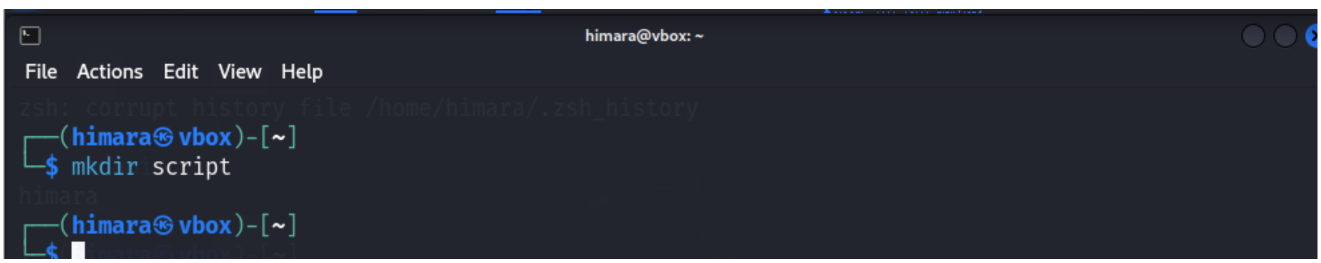
• Function

himara@vbox: ~/SNP_Lab
File Actions Edit View Help
(himara㉿vbox)-[~/SNP_Lab]
$ nano function_example.sh
(himara㉿vbox)-[~/SNP_Lab]
$ chmod +x function_example.sh
(himara㉿vbox)-[~/SNP_Lab]
$ ./function_example.sh
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
(himara㉿vbox)-[~/SNP_Lab]
$



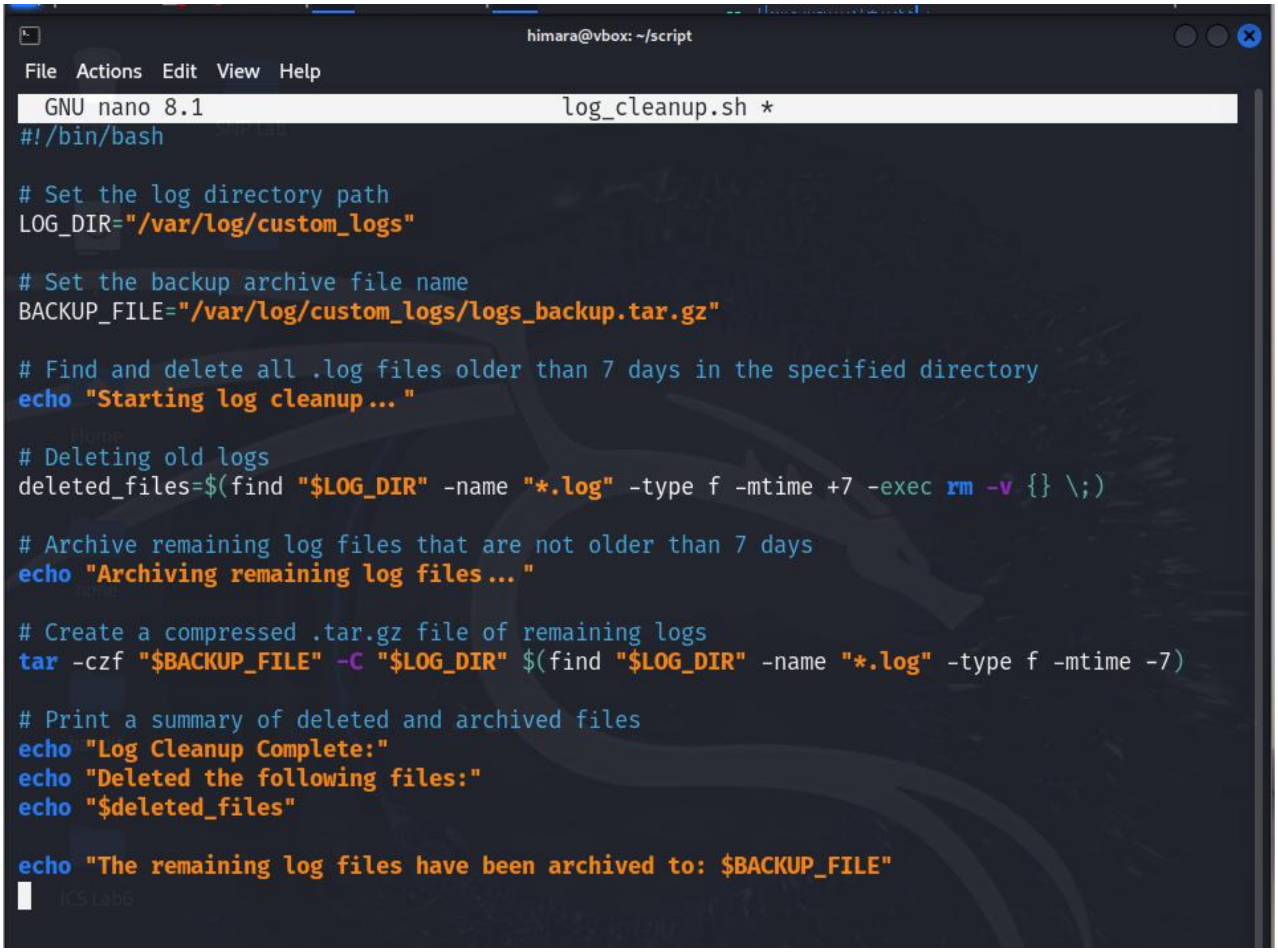
Task 01

1. Create a new directory in /hime/himara with the name **script**.



2. You can use a text editor **nano** to create the script. You can create a nano log_cleanup.sh text file.

3. Write the script



- **LOG_DIR="/var/log/custom_logs"** = Set the log directory path.
- **BACKUP_FILE="/var/log/custom_logs/logs_backup.tar.gz"** = Set the backup archive file name.
- **echo "Starting log cleanup..."** = Find and delete all .log files older than 7 days in the specified directory.
- **deleted_files=$(find "$LOG_DIR" -name "*.log" -type f -mtime +7 -exec rm -v {} \;)** = Deleting old logs.
- **echo "Archiving remaining log files..."** = Archive remaining log files that are not older than 7 days.
- **tar -czf "$BACKUP_FILE" -C "$LOG_DIR" $(find "$LOG_DIR" -name "*.log" -type f -mtime -7)** = Create a compressed .tar.gz file of remaining logs.

- **echo "Log Cleanup Complete:"**
- **echo "Deleted the following files:"**
- **echo "$deleted_files"**
- **echo "The remaining log files have been archived to: $BACKUP_FILE"**

print a summary of deleted and archived files.

4. Make the script executable

To execute the script, you should provide it with executable permissions. Run the following command in the terminal.This will render the script executable.

```
┌──(himara㉿vbox)-[~/script]
└─$ chmod +x log_cleanup.sh
```

5. Test the Script

```
┌──(himara㉿vbox)-[~/script]
└─$ ./log_cleanup.sh

Starting log cleanup...
Archiving remaining log files...
tar: Cowardly refusing to create an empty archive
Try 'tar --help' or 'tar --usage' for more information.
Log Cleanup Complete:
Deleted the following files:

The remaining log files have been archived to: /var/log/custom_logs/logs_backup.tar.gz
```

6. Set Up a Cron Job

You must add this script to your cron tasks in order to have it run automatically every Sunday at midnight. Open the crontab file by **“crontab -e”** running thi code and add the following line to run the script every Sunday at 12:00 AM (midnight).

```
┌──(himara㉿vbox)-[~/script]
└─$ crontab -e

no crontab for himara - using an empty one
crontab: installing new crontab
```

```
himara@vbox: ~/script
File  Actions  Edit  View  Help
  GNU nano 8.1                    /tmp/crontab.8mNnwU/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command

0 0 * * SUN /home/himara/log_cleanup.sh >> /home/himara/log_cleanup.log 2>&1
```

Task 02

SSH (Secure Shell):

1. Update and Install the SSH server package

```
┌──(himara㉿vbox)-[~]
└─$ sudo apt update

[sudo] password for himara:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
2135 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

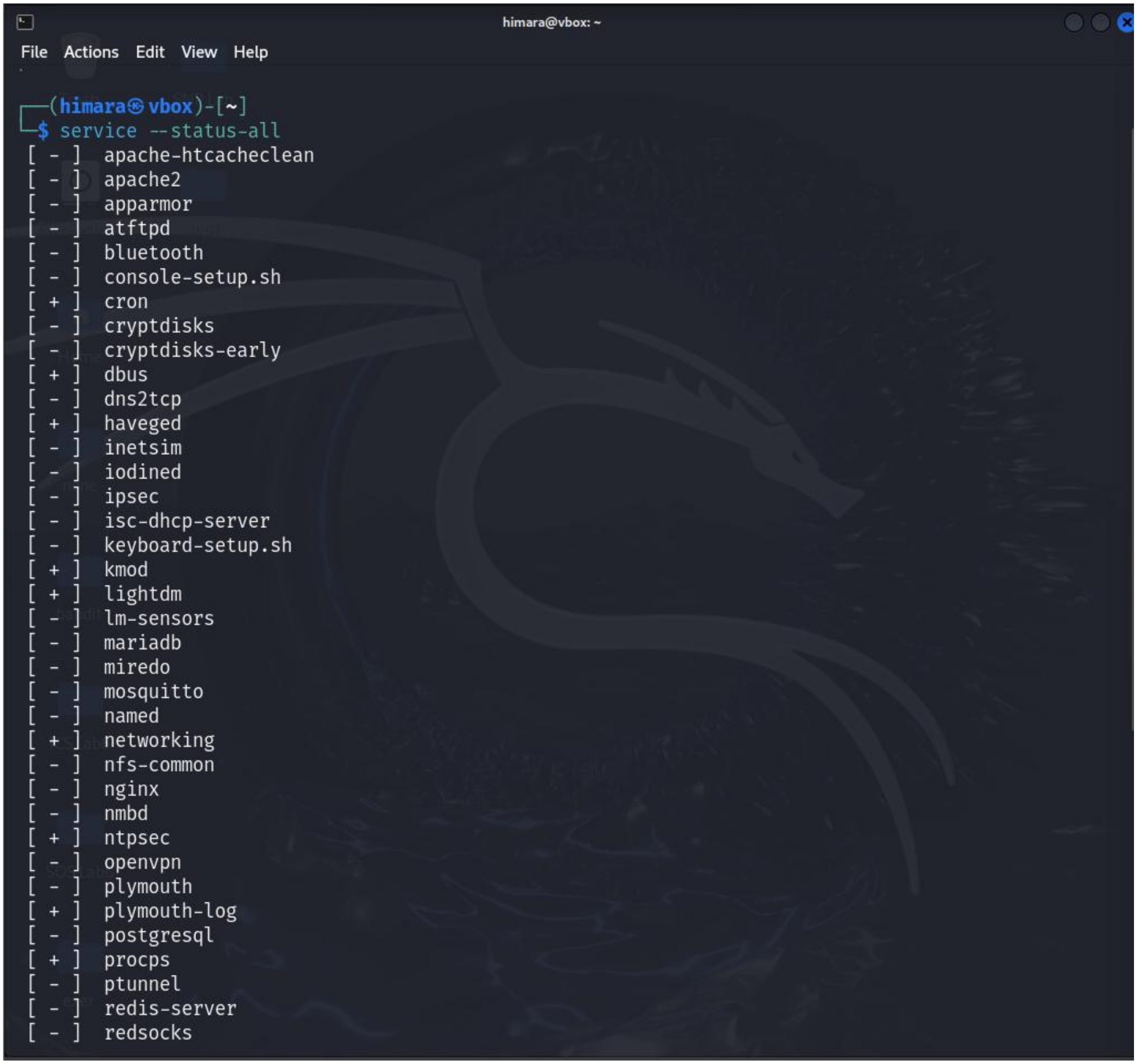
```
┌──(himara㉿vbox)-[~]
└─$ sudo apt install openssh-server

openssh-server is already the newest version (1:9.9p2-2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2135
```

2. Verify the SSH service is running

Check whether if thw SSH service is running properly.

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl status ssh

● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: disabled)
     Active: active (running) since Sun 2025-05-04 09:51:18 EDT; 3min 52s ago
 Invocation: ed1b46a8df2e4bc0923f7ffa28cb597e
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 838 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 857 (sshd)
      Tasks: 1 (limit: 4548)
     Memory: 2.1M (peak: 2.3M)
        CPU: 92ms
     CGroup: /system.slice/ssh.service
             └─857 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 04 09:51:18 vbox systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 04 09:51:18 vbox sshd[857]: Server listening on 0.0.0.0 port 22.
May 04 09:51:18 vbox systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
May 04 09:51:18 vbox sshd[857]: Server listening on :: port 22.
```

- After installing and enabling SSH, now we check the status of all available services on your system.
- It provides a quick overview of whether services are running, stopped, or inactive.
- According to the below image the SSH service now active, which is shown by ‘ +‘ mark.



```
[ - ]  rpcbind
[ - ]  rsync
[ - ]  rwhod
[ - ]  saned
[ - ]  screen-cleanup
[ - ]  selinux-autorelabel
[ - ]  smartmontools
[ - ]  smbd
[ - ]  snmpd
[ - ]  speech-dispatcher
[ - ]  ssh
[ - ]  sslh
[ - ]  stunnel4
[ - ]  sudo
[ - ]  sysstat
[ + ]  ufw
[ + ]  virtualbox-guest-utils
[ - ]  x11-common

┌──(himara㉿vbox)-[~]
└─$ 
```

3. Enable SSH service to start on boot

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl enable ssh

Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sy
sv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
```

4. Edit the SSH configuration file

- The SSH configuration file is located at /etc/ssh/sshd_config.

```
┌──(himara㉿vbox)-[~]
└─$ sudo nano /etc/ssh/sshd_config

┌──(himara㉿vbox)-[~]
└─$ 
```

After accessing the this file you can do the following modifications:

- Port = 2255
- PermitRootLogin = no
- PasswordAuthentication = Set this to 'no' if you want to enforce key based authentication.
- AllowUsers: Specify which users are allowed to connect via SSH.

```
himara@vbox: ~
File  Actions  Edit  View  Help
  GNU nano 8.1                         /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file.  See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

Port 2255
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10
                                          [ Read 122 lines ]
^G Help       ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo
```

5. Restart the service

After making this changes restart the SSH service.

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl restart ssh

┌──(himara㉿vbox)-[~]
└─$ 
```

6. Allow SSH through the firewall

```
┌──(himara㉿vbox)-[~]
└─$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)

┌──(himara㉿vbox)-[~]
└─$ sudo ufw allow 2255/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)

┌──(himara㉿vbox)-[~]
└─$ sudo ufw enable
Firewall is active and enabled on system startup

┌──(himara㉿vbox)-[~]
└─$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
2255/tcp                   ALLOW       Anywhere
67/udp                     ALLOW       Anywhere
53/udp                     ALLOW       Anywhere
53/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
2255/tcp (v6)              ALLOW       Anywhere (v6)
67/udp (v6)                ALLOW       Anywhere (v6)
53/udp (v6)                ALLOW       Anywhere (v6)
53/tcp (v6)                ALLOW       Anywhere (v6)
```

7. Locating the IP address of the server computer

```
┌──(himara㉿vbox)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.103  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::a00:27ff:fe0b:111a  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:0b:11:1a  txqueuelen 1000  (Ethernet)
        RX packets 8  bytes 2634 (2.5 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 3520 (3.4 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 14  bytes 780 (780.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 14  bytes 780 (780.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

8. Connect to the Linux Machine from Another Computer Using SSH

Open a terminal on the remote machine and run the following command:

```
┌──(himara㉿vbox)-[~]
└─$ ssh himara@192.168.56.103 -p 2255
Linux vbox 6.8.11-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.8.11-1kali2 (2024-05-30) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat May  3 16:06:56 2025 from 192.168.56.103
zsh: corrupt history file /home/himara/.zsh_history
```

Task 03

iptables and ACLs

1. Find the IP ranges for Social Media Websites
   - Facebook: 157.240.0.0/16 (primary IP range for Facebook)
   - Instagram: 31.13.64.0/18 (Instagram is owned by Facebook, so the IP ranges overlap)
   - Twitter: 199.59.148.0/22 (Twitter's IP range)

2. Add iptables Rules to Block Social Media Websites

Open your VM terminal and run the following iptables command to block the sociam media IP.

```
himara@vbox: ~
File  Actions  Edit  View  Help
zsh: corrupt history file /home/himara/.zsh_history
┌──(himara㉿vbox)-[~]
└─$ # Block Facebook IP range
sudo iptables -A OUTPUT -d 157.240.0.0/16 -j REJECT   # Facebook

# Block Instagram IP range (same as Facebook's IP ranges)
sudo iptables -A OUTPUT -d 31.13.64.0/18 -j REJECT    # Instagram

# Block Twitter IP range
sudo iptables -A OUTPUT -d 199.59.148.0/22 -j REJECT   # Twitter

[sudo] password for himara:

┌──(himara㉿vbox)-[~]
└─$ 
```

3. Save the iptables Rules

```
┌──(himara㉿vbox)-[~]
└─$ sudo iptables-save > /etc/iptables/rules.v4
```

4. Allow Only Secure Web Browsing (HTTPS)

Block HTTP (Port 80) Traffic

```
┌──(himara㉿vbox)-[~]
└─$ sudo iptables -A OUTPUT -p tcp --dport 80 -j REJECT   # Block HTTP (port 80)
```

Allow HTTPS (Port 443) Traffic

```
┌──(himara㉿vbox)-[~]
└─$ sudo iptables -A OUTPUT -p tcp --dport 443 -j ACCEPT   # Allow HTTPS (port 443)
```

5. Install of the iptables persistent

```
┌──(himara㉿vbox)-[~]
└─$ sudo apt install iptables-persistent

iptables-persistent is already the newest version (1.0.23).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2123
```

6. Save the iptables Rules

```
┌──(himara㉿vbox)-[~]
└─$ sudo iptables-save > /etc/iptables/rules.v4
```

7. Verify the Configuration

```
┌──(himara㉿vbox)-[~]
└─$ sudo iptables -L

Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
REJECT     all  --  anywhere             157.240.0.0/16       reject-with icmp-port-unreachable
REJECT     all  --  anywhere             31.13.64.0/18        reject-with icmp-port-unreachable
REJECT     all  --  anywhere             199.59.148.0/22      reject-with icmp-port-unreachable
REJECT     tcp  --  anywhere             anywhere             tcp dpt:http reject-with icmp-port-unreachable
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:https
```

Task 04

Web Server (Apache/Nginx)

1. Install Apache2

```
┌──(himara㉿vbox)-[~]
└─$ sudo apt-get install apache2
[sudo] password for himara:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.63-1).
apache2 set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 2127 not upgraded.
```

2. Start the apache2

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl start apache2
sudo systemctl enable apache2

Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

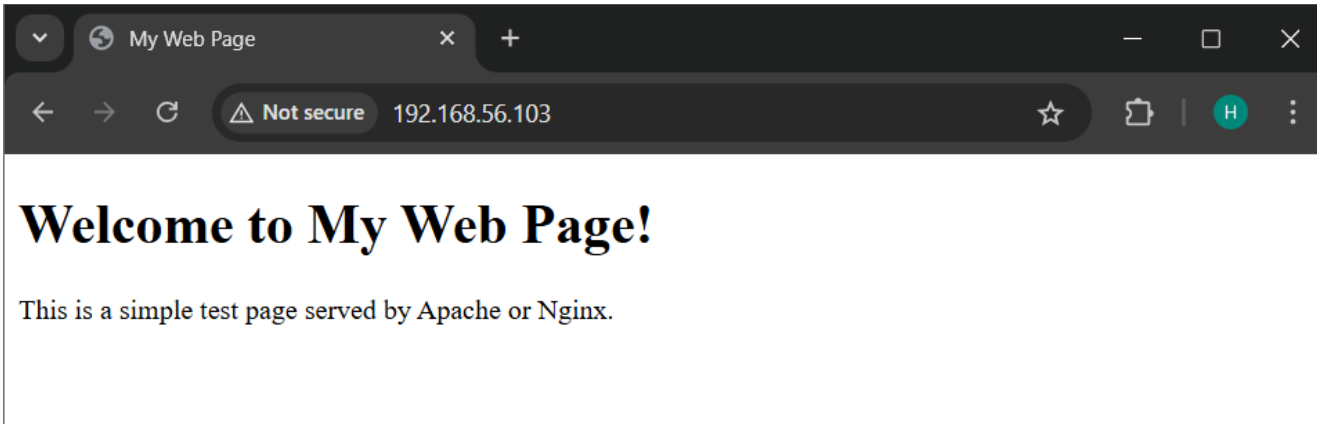
3. Check if Apache is running

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: disabled)
     Active: active (running) since Tue 2025-05-06 12:58:17 EDT; 1h 4min ago
 Invocation: 4485e797a54945ea861bd5018cc4fb3c
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 857 (apache2)
      Tasks: 6 (limit: 4548)
     Memory: 20.6M (peak: 21.1M)
        CPU: 809ms
     CGroup: /system.slice/apache2.service
             ├─857 /usr/sbin/apache2 -k start
             ├─861 /usr/sbin/apache2 -k start
             ├─862 /usr/sbin/apache2 -k start
             ├─863 /usr/sbin/apache2 -k start
             ├─864 /usr/sbin/apache2 -k start
             └─865 /usr/sbin/apache2 -k start

May 06 12:58:17 vbox systemd[1]: Starting apache2.service - The Apache HTTP Server...
May 06 12:58:17 vbox systemd[1]: Started apache2.service - The Apache HTTP Server.
```

4. Create an **index.html** file in the web server's root directory

```
himara@vbox: ~                                   CPU usage: 23.7%
File  Actions  Edit  View  Help
  GNU nano 8.1                        /var/www/html/index.html

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Web Page</title>
</head>
<body>
    <h1>Welcome to My Web Page!</h1>
    <p>This is a simple test page served by Apache or Nginx.</p>
</body>
</html>
```

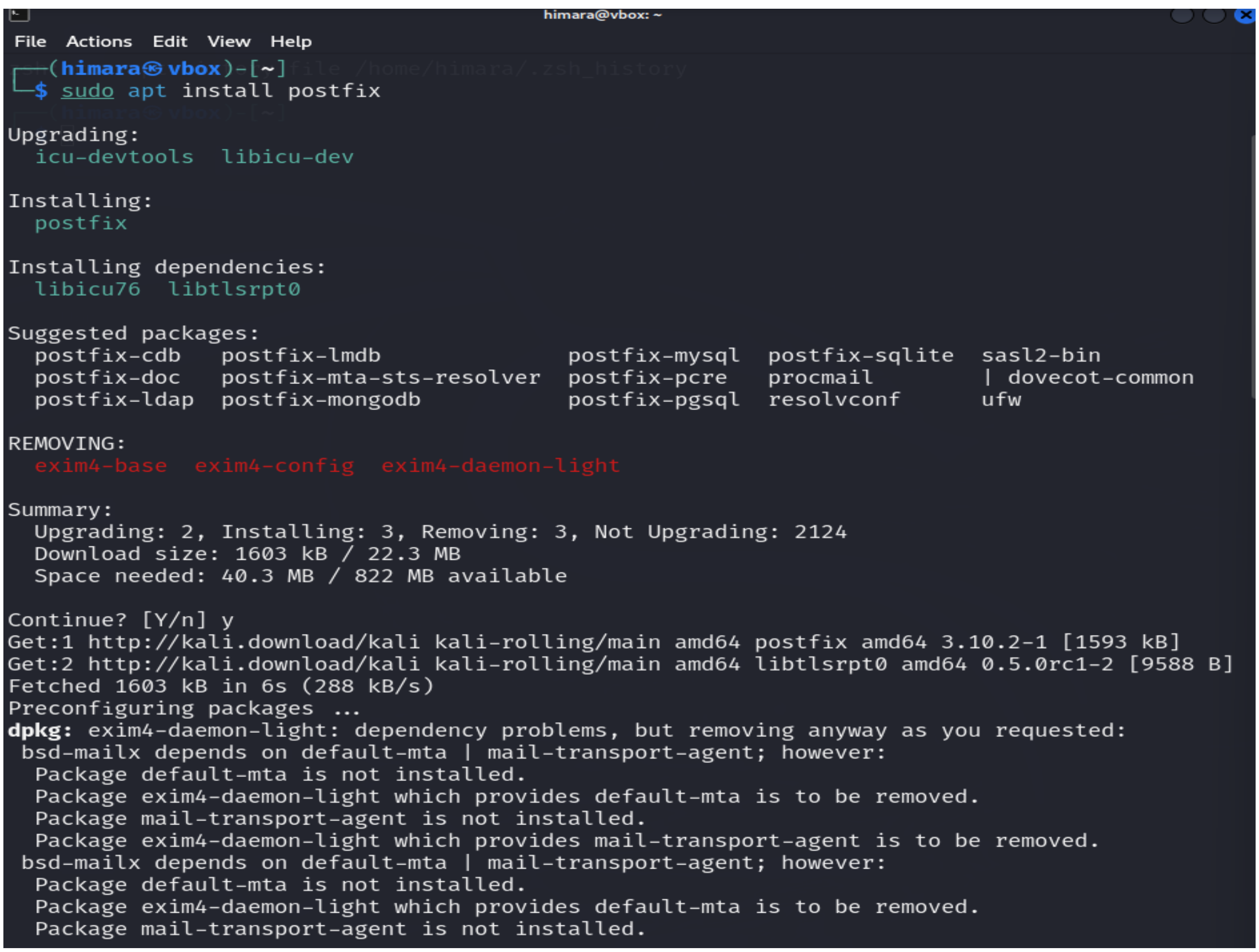
5. Access the page from another machine



Task 05

Email Server (postfix/Exim)

1. Install Postfix



```
himara@vbox: ~
File  Actions  Edit  View  Help
┌──(himara㉿vbox)-[~]
└─$ sudo apt install postfix

Upgrading:
  icu-devtools  libicu-dev

Installing:
  postfix

Installing dependencies:
  libicu76  libtlsrpt0

Suggested packages:
  postfix-cdb   postfix-lmdb                postfix-mysql  postfix-sqlite  sasl2-bin
  postfix-doc   postfix-mta-sts-resolver    postfix-pcre   procmail        | dovecot-common
  postfix-ldap  postfix-mongodb             postfix-pgsql  resolvconf      ufw

REMOVING:
  exim4-base  exim4-config  exim4-daemon-light

Summary:
  Upgrading: 2, Installing: 3, Removing: 3, Not Upgrading: 2124
  Download size: 1603 kB / 22.3 MB
  Space needed: 40.3 MB / 822 MB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 postfix amd64 3.10.2-1 [1593 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 libtlsrpt0 amd64 0.5.0rc1-2 [9588 B]
Fetched 1603 kB in 6s (288 kB/s)
Preconfiguring packages ...
dpkg: exim4-daemon-light: dependency problems, but removing anyway as you requested:
 bsd-mailx depends on default-mta | mail-transport-agent; however:
  Package default-mta is not installed.
  Package exim4-daemon-light which provides default-mta is to be removed.
  Package mail-transport-agent is not installed.
  Package exim4-daemon-light which provides mail-transport-agent is to be removed.
 bsd-mailx depends on default-mta | mail-transport-agent; however:
  Package default-mta is not installed.
  Package exim4-daemon-light which provides default-mta is to be removed.
  Package mail-transport-agent is not installed.
```

Configure Postfix:

2. Edit the Postfix configuration file

```
┌──(himara㉿vbox)-[~]
└─$ sudo nano /etc/postfix/main.cf
```

```
himara@vbox: ~
File  Actions  Edit  View  Help
  GNU nano 8.1                          /etc/postfix/main.cf *
# See /usr/share/postfix/main.cf.dist for a commented, more complete version

myhostname = vbox.kali
mydomain = kali.local
mydestination = $myhostname, localhost.$mydomain, localhost
inet_protocols = all

# See http://www.postfix.org/COMPATIBILITY_README.html
compatibility_level = 3.9

# Which domain that locally-originated mail appears to come from.
# Debian policy suggests to read this value from /etc/mailname.
#XX needs a review in postinst&config
#myorigin = /etc/mailname
#myorigin = $mydomain
myorigin = /etc/mailname

# Text that follows the 220 code in the SMTP server's greeting banner.
# You MUST specify $myhostname at the start due to an RFC requirement.
smtpd_banner = $myhostname ESMTP $mail_name (Debian)

# IP protocols to use: ipv4, ipv6, or all
# (set this explicitly so `post-install upgrade-configuration' wont complain)
inet_protocols = all

# List of "trusted" SMTP clients (maptype:mapname allowed) that have more
# privileges than "strangers".  If mynetworks is not specified (the default),
# mynetworks_style is used to compute its value.
#mynetworks_style = class
#mynetworks_style = subnet
mynetworks_style = host
#
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128

# Uncomment the next line to generate "delayed mail" warnings
#delay_warning_time = 4h

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

3. Restart Postfix

```
┌──(himara㉿vbox)-[~]
└─$ sudo systemctl restart postfix
```

4. Test Postfix Installation

Install the mail command

```
himara@vbox: ~
File  Actions  Edit  View  Help
zsh: corrupt history file /home/himara/.zsh_history
┌──(himara㉿vbox)-[~]
└─$ sudo apt install mailutils

Upgrading:
  libltdl7

Installing:
  mailutils

Installing dependencies:
  gsasl-common    libgsasl18    libmailutils9t64  mailutils-common
  guile-3.0-libs  libgssglue1   libntlm0

Suggested packages:
  mailutils-mh  mailutils-doc

Summary:
  Upgrading: 1, Installing: 8, Removing: 0, Not Upgrading: 2123
  Download size: 9388 kB / 9804 kB
  Space needed: 63.7 MB / 789 MB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 gsasl-common all 2.2.2-1 [52.6 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 guile-3.0-libs amd64 3.0.10+really3.
0.10-4 [6877 kB]
Get:4 http://mirror.aktkn.sg/kali kali-rolling/main amd64 libntlm0 amd64 1.8-4 [22.4 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libgssglue1 amd64 0.9-1+b1 [20.8 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 libgsasl18 amd64 2.2.2-1 [80.0 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 mailutils-common all 1:3.19-1 [818 k
B]
Get:7 http://kali.download/kali kali-rolling/main amd64 libmailutils9t64 amd64 1:3.19-1 [943
 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 mailutils amd64 1:3.19-1 [574 kB]
Fetched 9388 kB in 1min 7s (140 kB/s)
Selecting previously unselected package gsasl-common.
(Reading database ... 397051 files and directories currently installed.)
Preparing to unpack .../0-gsasl-common_2.2.2-1_all.deb ...
Unpacking gsasl-common (2.2.2-1) ...
Selecting previously unselected package guile-3.0-libs:amd64.
```

Send a test emai

```
┌──(himara㉿vbox)-[~]
└─$ echo "This is a test email" | mail -s "Test Email" vbox.kali
```

4.Linux GDB

1. Select the Appropriate Executable

Check the system architecture.

```
┌──(himara㉿vbox)-[~]
└─$ uname -m

x86_64
```

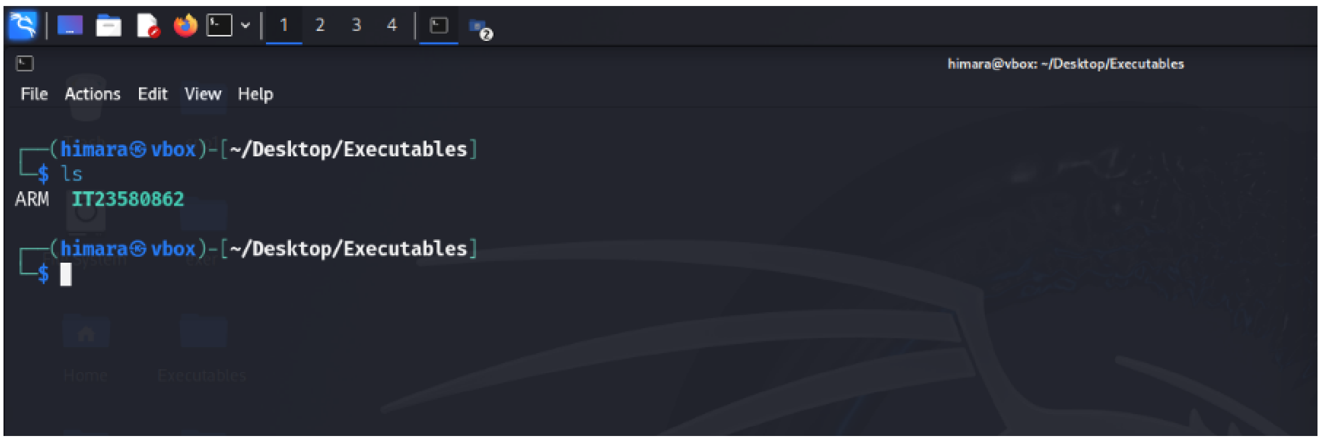
- Based on the output, select the correct executable:
- If the output is x86_64, use the 64-bit executable.
- If the output is i686, use the 32-bit executable.
- If the output is arm, use the ARM-based executable.

2. Install the GDB

```
himara@vbox: ~/Desktop/Executables
File  Actions  Edit  View  Help

┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$ sudo apt install gdb
Upgrading:
  gdb

Summary:
  Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 2122
  Download size: 3992 kB
  Space needed: 0 B / 721 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 gdb amd64 16.3-1 [3992 kB]
Fetched 3992 kB in 3s (1566 kB/s)
```

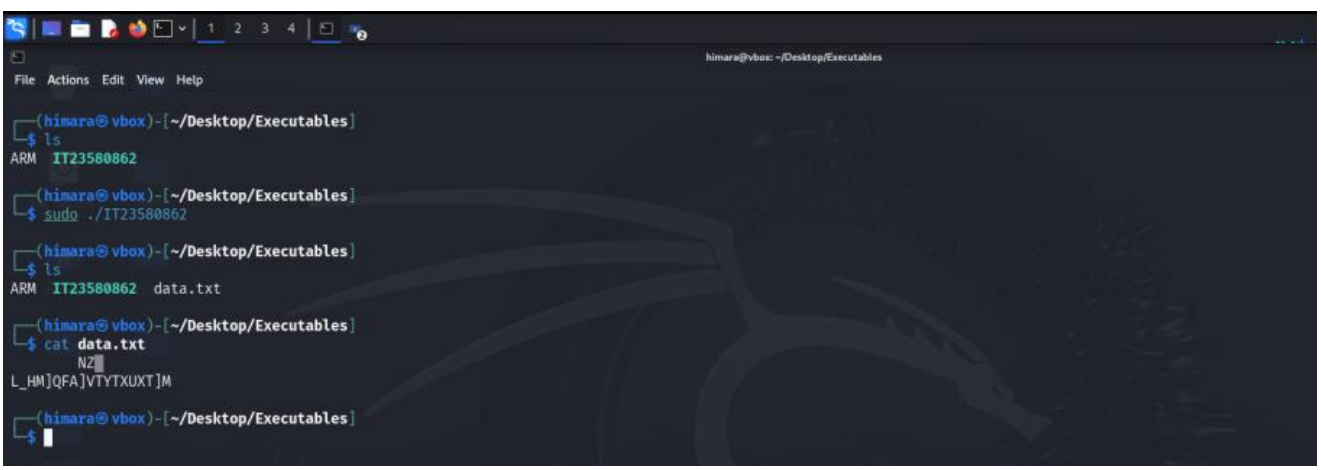
3. Execute the file

4. Change the file



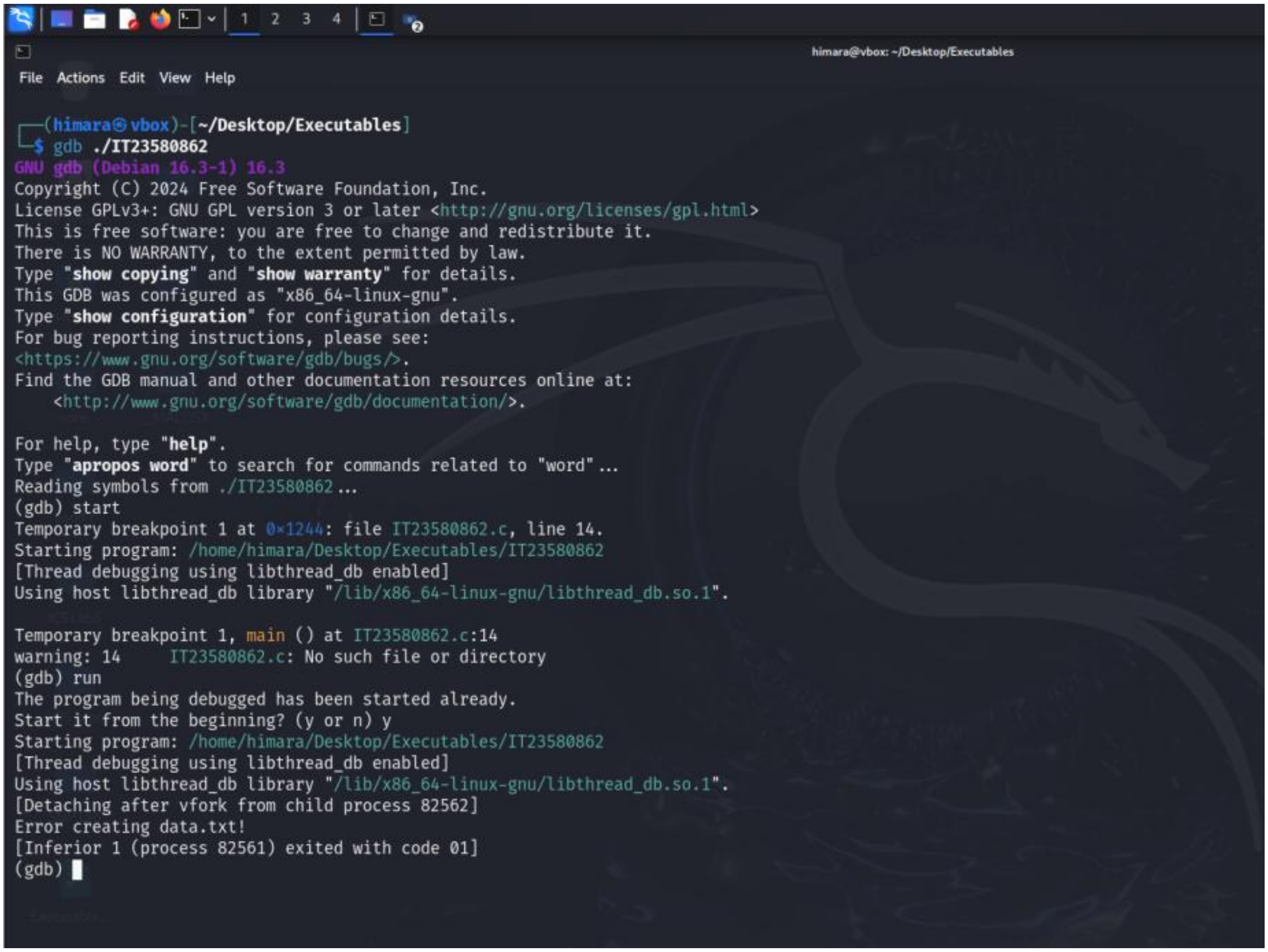
5. Analyze the generated executable

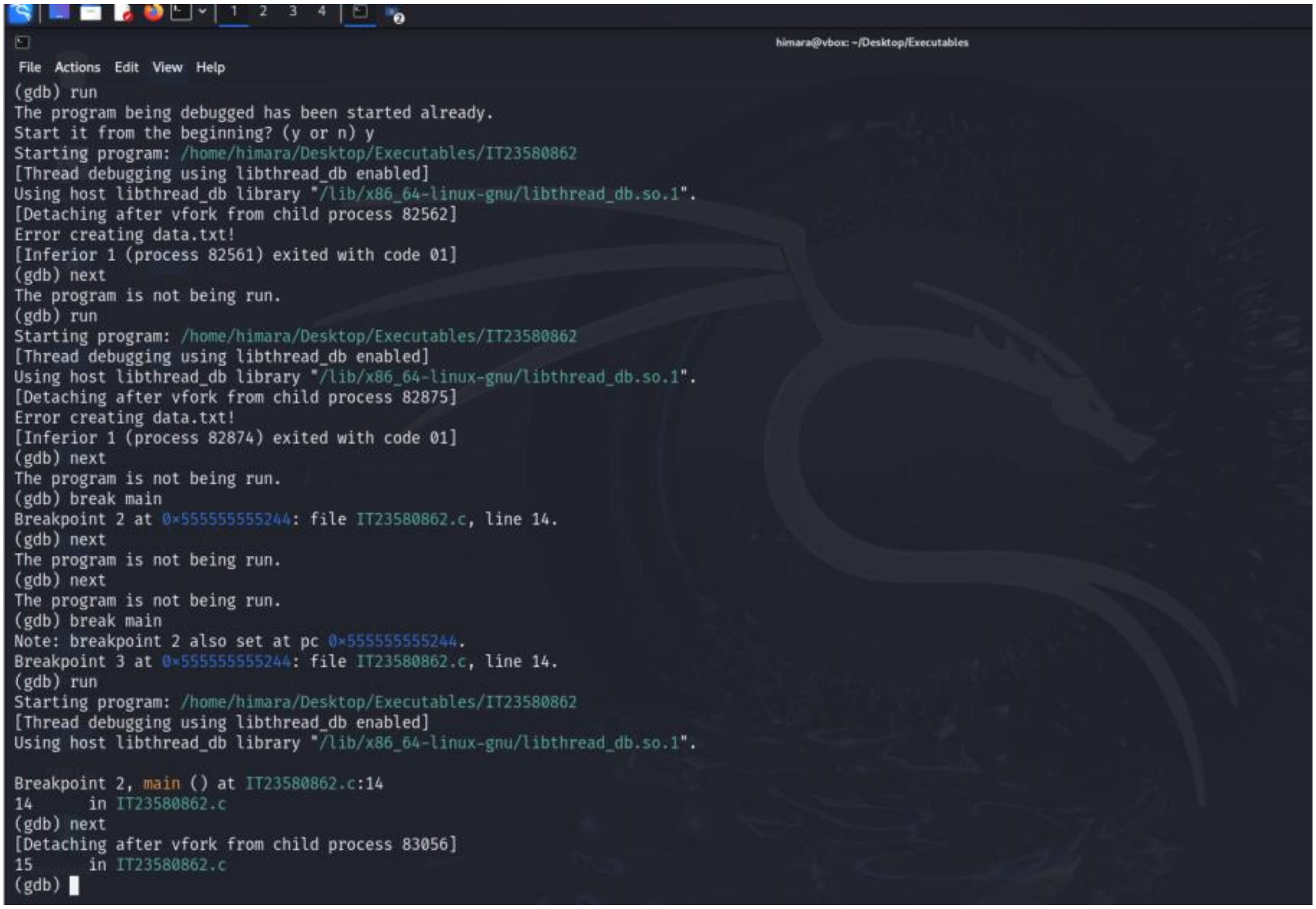
Execute the newly generated executable and analyze the output.

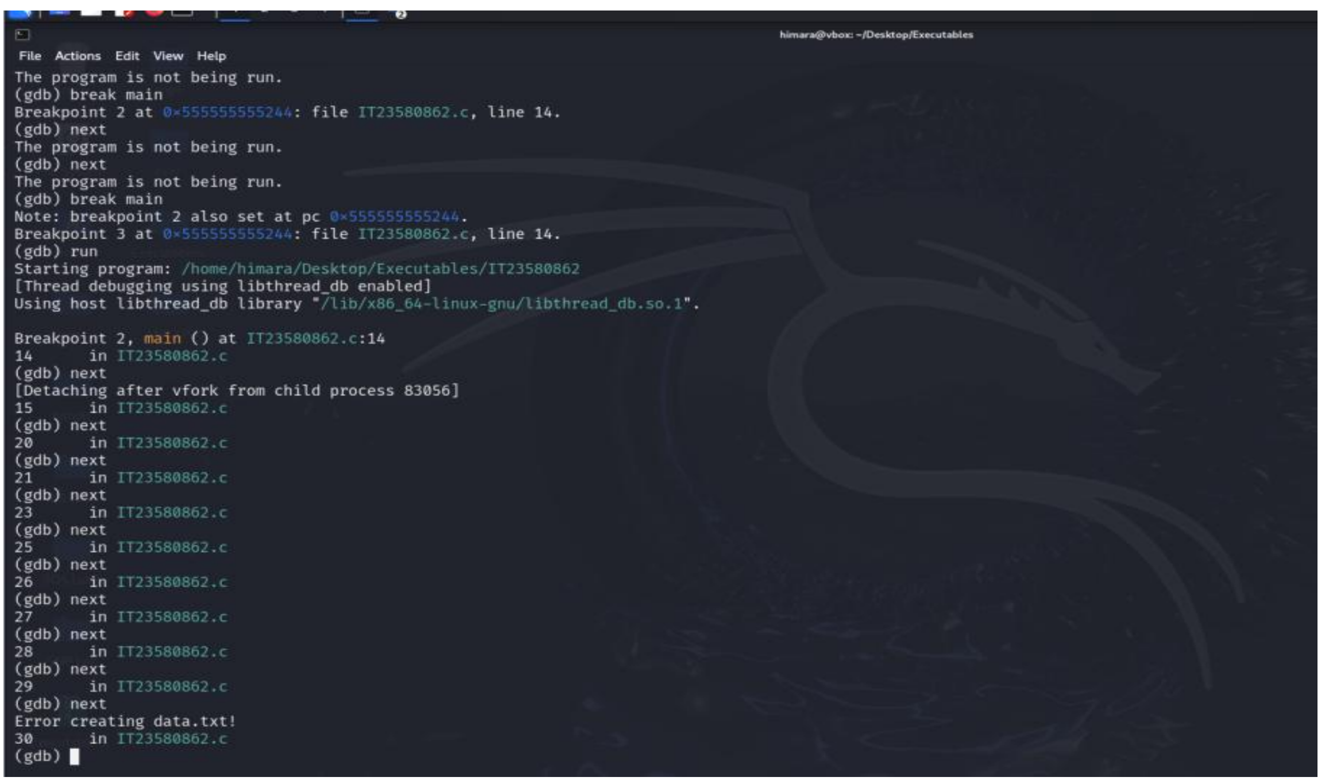


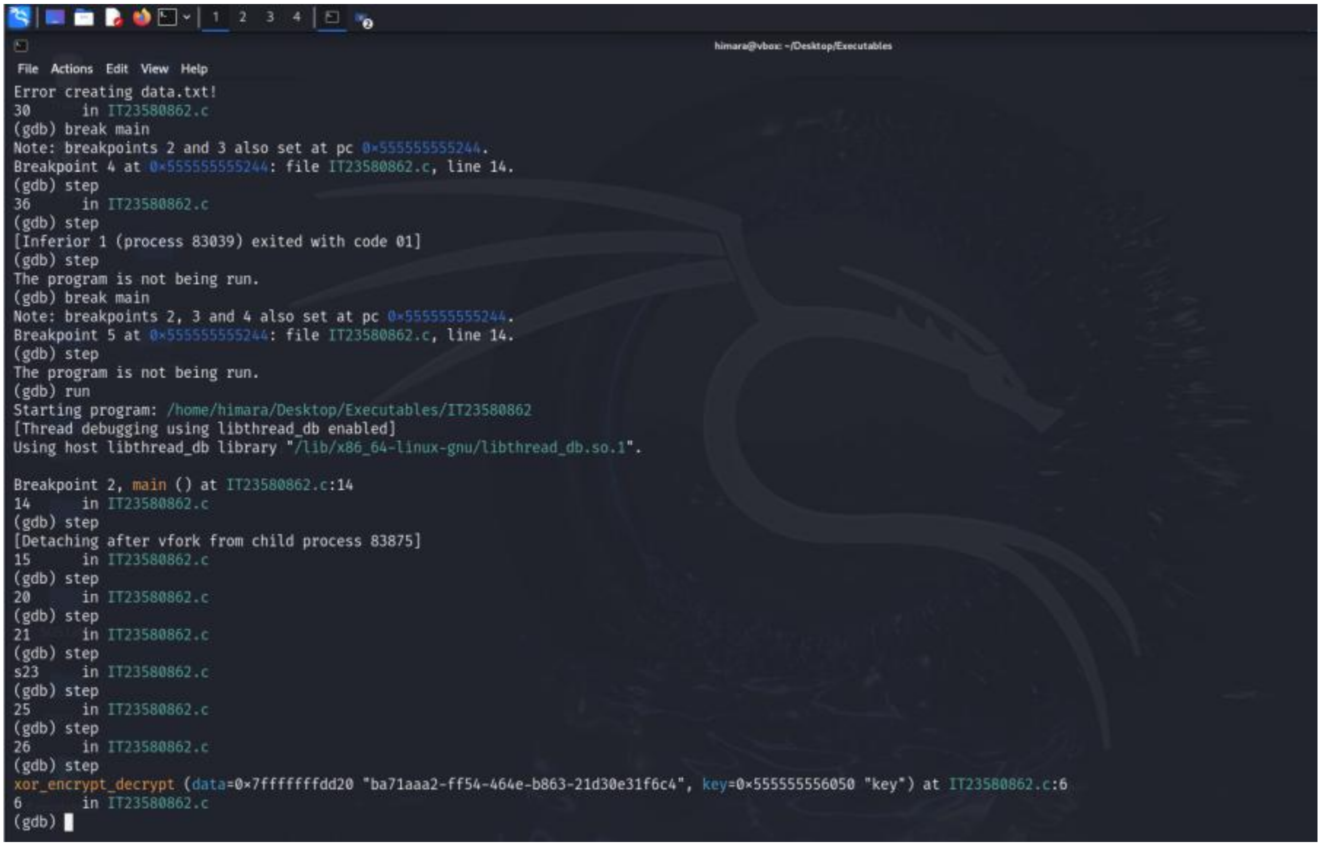
6. Debug the New Executable Using GDB

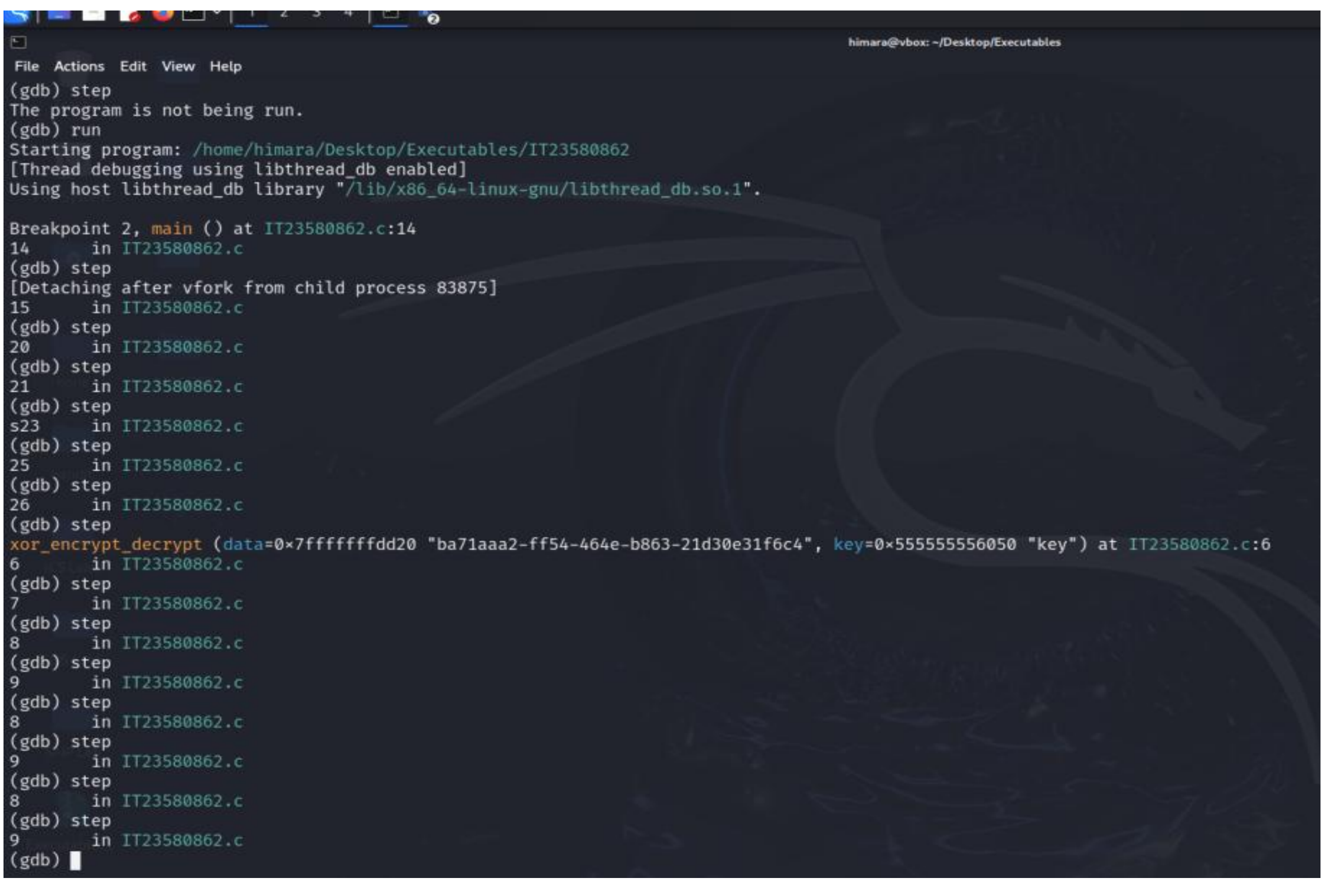
Start the debugging analyze the program flow.



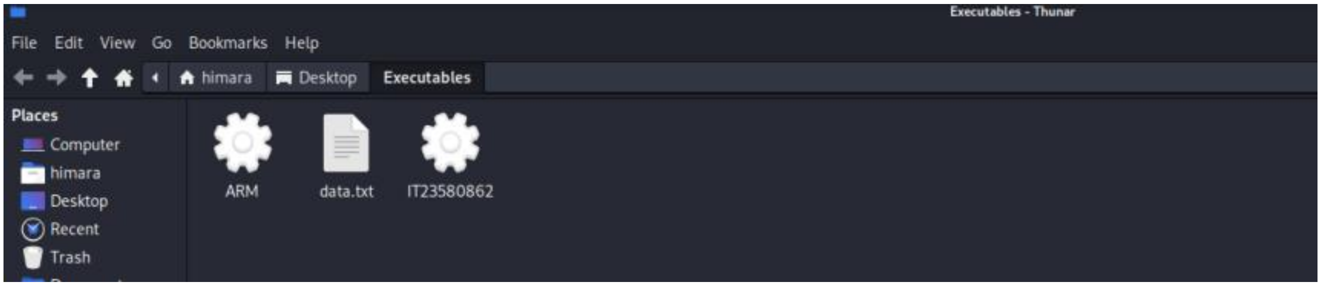

himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
(gdb) run
The program being debugged has been started already.
Start it from the beginning? (y or n) y
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[Detaching after vfork from child process 82562]
Error creating data.txt!
[Inferior 1 (process 82561) exited with code 01]
(gdb) next
The program is not being run.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
[Detaching after vfork from child process 82875]
Error creating data.txt!
[Inferior 1 (process 82874) exited with code 01]
(gdb) next
The program is not being run.
(gdb) break main
Breakpoint 2 at 0x555555555244: file IT23580862.c, line 14.
(gdb) next
The program is not being run.
(gdb) next
The program is not being run.
(gdb) break main
Note: breakpoint 2 also set at pc 0x555555555244.
Breakpoint 3 at 0x555555555244: file IT23580862.c, line 14.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Breakpoint 2, main () at IT23580862.c:14
14 in IT23580862.c
(gdb) next
[Detaching after vfork from child process 83056]
15 in IT23580862.c
(gdb)


himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
The program is not being run.
(gdb) break main
Breakpoint 2 at 0x555555555244: file IT23580862.c, line 14.
(gdb) next
The program is not being run.
(gdb) next
The program is not being run.
(gdb) break main
Note: breakpoint 2 also set at pc 0x555555555244.
Breakpoint 3 at 0x555555555244: file IT23580862.c, line 14.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Breakpoint 2, main () at IT23580862.c:14
14 in IT23580862.c
(gdb) next
[Detaching after vfork from child process 83056]
15 in IT23580862.c
(gdb) next
20 in IT23580862.c
(gdb) next
21 in IT23580862.c
(gdb) next
23 in IT23580862.c
(gdb) next
25 in IT23580862.c
(gdb) next
26 in IT23580862.c
(gdb) next
27 in IT23580862.c
(gdb) next
28 in IT23580862.c
(gdb) next
29 in IT23580862.c
(gdb) next
Error creating data.txt!
30 in IT23580862.c
(gdb)


1 2 3 4
himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
Error creating data.txt!
30 in IT23580862.c
(gdb) break main
Note: breakpoints 2 and 3 also set at pc 0x555555555244.
Breakpoint 4 at 0x555555555244: file IT23580862.c, line 14.
(gdb) step
36 in IT23580862.c
(gdb) step
[Inferior 1 (process 83039) exited with code 01]
(gdb) step
The program is not being run.
(gdb) break main
Note: breakpoints 2, 3 and 4 also set at pc 0x555555555244.
Breakpoint 5 at 0x555555555244: file IT23580862.c, line 14.
(gdb) step
The program is not being run.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Breakpoint 2, main () at IT23580862.c:14
14 in IT23580862.c
(gdb) step
[Detaching after vfork from child process 83875]
15 in IT23580862.c
(gdb) step
20 in IT23580862.c
(gdb) step
21 in IT23580862.c
(gdb) step
s23 in IT23580862.c
(gdb) step
25 in IT23580862.c
(gdb) step
26 in IT23580862.c
(gdb) step
xor_encrypt_decrypt (data=0x7fffffffdd20 "ba71aaa2-ff54-464e-b863-21d30e31f6c4", key=0x555555556050 "key") at IT23580862.c:6
6 in IT23580862.c
(gdb)


himara@vbox: ~/Desktop/Executables
File Actions Edit View Help
(gdb) step
The program is not being run.
(gdb) run
Starting program: /home/himara/Desktop/Executables/IT23580862
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".
Breakpoint 2, main () at IT23580862.c:14
14 in IT23580862.c
(gdb) step
[Detaching after vfork from child process 83875]
15 in IT23580862.c
(gdb) step
20 in IT23580862.c
(gdb) step
21 in IT23580862.c
(gdb) step
s23 in IT23580862.c
(gdb) step
25 in IT23580862.c
(gdb) step
26 in IT23580862.c
(gdb) step
xor_encrypt_decrypt (data=0x7fffffffdd20 "ba71aaa2-ff54-464e-b863-21d30e31f6c4", key=0x555555556050 "key") at IT23580862.c:6
6 in IT23580862.c
(gdb) step
7 in IT23580862.c
(gdb) step
8 in IT23580862.c
(gdb) step
9 in IT23580862.c
(gdb) step
8 in IT23580862.c
(gdb) step
9 in IT23580862.c
(gdb) step
8 in IT23580862.c
(gdb) step
9 in IT23580862.c
(gdb)

7. Flie system changes when debugging



8. Convert the executable file to a readable format

- Use the command: *strings ./IT23580862*. It will show the source code

```
┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$ strings ./IT23580862
/lib64/ld-linux-x86-64.so.2
__cxa_finalize
fgets
strcspn
__libc_start_main
fputs
fopen
fclose
pclose
strlen
popen
libc.so.6
GLIBC_2.2.5
GLIBC_2.34
_ITM_deregisterTMCloneTable
__gmon_start__
_ITM_registerTMCloneTable
PTE1
u+UH
sudo cat /sys/class/dmi/id/product_uuid
Error retrieving data
data.txt
Error creating data.txt!
;*3$"
GCC: (Debian 13.3.0-3) 13.3.0
__off_t
_IO_read_ptr
_chain
size_t
_shortbuf
_IO_buf_base
long long unsigned int
GNU C17 13.3.0 -mtune=generic -march=x86-64 -g -fasynchronous-unwind-tables
long long int
_fileno
_IO_read_end
_flags
```

himara@vbox: ~/Desktop/Executables

File Actions Edit View Help

```
_freeres_list
main
_IO_write_base
IT23580862.c
/home/himara/Desktop/Executables
/usr/lib/gcc/x86_64-linux-gnu/13/include
/usr/include/x86_64-linux-gnu/bits
/usr/include/x86_64-linux-gnu/bits/types
/usr/include
stddef.h
types.h
struct_FILE.h
stdio.h
string.h
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
void xor_encrypt_decrypt(char *data, const char *key) {
    size_t data_len = strlen(data);
    size_t key_len = strlen(key);
    for (size_t i = 0; i < data_len; i++) {
        data[i] ^= key[i % key_len];
    }
int main() {
    FILE *fp = popen("sudo cat /sys/class/dmi/id/product_uuid", "r");
    if (fp == NULL) {
        printf("Error retrieving data\n");
        return 1;
    }
    char uuid[50];
    fgets(uuid, sizeof(uuid), fp);
    pclose(fp);
    uuid[strcspn(uuid, "\n")] = 0; // Remove newline
    const char *key = "key";
    xor_encrypt_decrypt(uuid, key);
    FILE *out = fopen("data.txt", "w");
    if (out == NULL) {
        printf("Error creating data.txt!\n");
        return 1;
    }
```

```
  FILE *out = fopen("data.txt", "w");
  if (out == NULL) {
      printf("Error creating data.txt!\n");
      return 1;
  }
  fprintf(out, "%s", uuid);
  fclose(out);
  return 0;
Scrt1.o
__abi_tag
crtstuff.c
deregister_tm_clones
__do_global_dtors_aux
completed.0
__do_global_dtors_aux_fini_array_entry
frame_dummy
__frame_dummy_init_array_entry
IT23580862.c
__FRAME_END__
_DYNAMIC
__GNU_EH_FRAME_HDR
_GLOBAL_OFFSET_TABLE_
__libc_start_main@GLIBC_2.34
_ITM_deregisterTMCloneTable
_edata
fclose@GLIBC_2.2.5
_fini
strlen@GLIBC_2.2.5
pclose@GLIBC_2.2.5
fputs@GLIBC_2.2.5
strcspn@GLIBC_2.2.5
fgets@GLIBC_2.2.5
```

- Copy the C code and create the file using this command nano SNP.c and paste the code.

```
himara@vbox: ~/Desktop/Executables
File  Actions  Edit  View  Help
  GNU nano 8.1                              SNP.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
void xor_encrypt_decrypt(char *data, const char *key) {
    size_t data_len = strlen(data);
    size_t key_len = strlen(key);
    for (size_t i = 0; i < data_len; i++) {
        data[i] ^= key[i % key_len];
    }
int main() {
    FILE *fp = popen("sudo cat /sys/class/dmi/id/product_uuid", "r");
    if (fp == NULL) {
        printf("Error retrieving data\n");
        return 1;
    }
    char uuid[50];
    fgets(uuid, sizeof(uuid), fp);
    pclose(fp);
    uuid[strcspn(uuid, "\n")] = 0; // Remove newline
    const char *key = "key";
    xor_encrypt_decrypt(uuid, key);
    FILE *out = fopen("data.txt", "w");
    if (out == NULL) {
        printf("Error creating data.txt!\n");
        return 1;
    }
    fprintf(out, "%s", uuid);
    fclose(out);
    return 0;
}
```

Run the source code

```
┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$ nano SNP.c

┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$ sudo chmod +x SNP.c

┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$ sudo gcc SNP.c -o snp

┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$
```

9. Create a Python script to decode the data.txt

```
himara@vbox: ~/Desktop/Executables
File  Actions  Edit  View  Help
 GNU nano 8.1                              decrypt.py *
def xor_decrypt(data, key):
    decrypted = ''
    for i in range(len(data)):
        decrypted += chr(ord(data[i]) ^ ord(key[i % len(key)]))
    return decrypted

#read the encrypted data
with open('data.txt', 'r') as f:
    encrypted = f.read()

key = "key"
decrypted_uuid = xor_decrypt(encrypted, key)
print("Decrypted UUID:", decrypted_uuid)
```

- Using this comman **python3 decryot.py** you can decrypt the data.txt.

```
┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$ nano decrypt.py

┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$ python3 decrypt.py
Decrypted UUID: ba71aaa2-af54-464e-b863-21d30e31f6c4

┌──(himara㉿vbox)-[~/Desktop/Executables]
└─$
```

• During GDB debugging, we discovered that the source code has XOR encryption. After conducting additional string analysis, we discovered the entire code, transferred it to a C file, and executed it.

• We demonstrated that it was responsible for creating the data.txt.

• A custom Python code was used to decrypt Data.txt, and the result displayed the system UUID, which is the encrypted content.